Optimisation of Holistically-Nested Edge Detection: A Systematic Literature Review

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Abstract

Background: Holistically Nested Edge Detection (HED) has become one of the most widely adopted deep learning architectures for boundary detection, formulating edge prediction as an image-to-image problem using a multiscale side-output mechanism. More than a decade of subsequent research has produced a large number of architectural variants, loss function modifications, and domain-specific adaptations. However, these contributions have emerged without systematic synthesis, leaving practitioners without evidence-based guidance for selecting architectures suited to specific accuracy, latency, and domain constraints.

Objective: This systematic literature review (SLR) addresses this gap by synthesising the HED optimisation literature into a structured, reusable evidence base.

Methods: We incorporated the PICOC (Population, Intervention, Comparison, Outcome, Context) framework for scope definition and adhered to the PRISMA 2020 guidelines for reporting transparency. A total of 48 primary studies published between 2015 and June 2026 were identified through seven major academic databases and assessed using a 12-criterion weighted quality assessment instrument validated by the modified Delphi process.

Results: No single optimisation strategy dominated all evaluation criteria. Backbone replacement (ResNet-50) increases computational cost but delivers consistent accuracy gains (median +1.8 percentage points Optimal Dataset Scale, ODS). Lightweight architectures (LDC, DexiNed) achieve 3–4× speedups with modest accuracy trade-offs of 0.6–0.9 pp ODS. Attention mechanisms offer reliable but bounded gains (median +1.4 pp ODS) with minimal computational costs. Metaheuristic hybrids (Social Spider Optimisation, Black Widow Optimisation) showed the highest reported improvement potential among the reviewed studies, although based on a limited three-study evidence base without reported computational cost (ODS gains ranging from +1.8 to +2.6 pp across three contributing studies on BSDS500, median +1.8 pp); this finding requires cautious interpretation given the small number of contributing studies and the absence of statistical significance testing in source papers. Reproducibility is a critical concern: 58.3% of studies (28 of 48) rely on proprietary datasets that are not publicly available, and only 20.8% (10 of 48) release their source code. The 12-criterion quality assessment yielded an average normalised score of 69.9% across all 48 studies, indicating acceptable but uneven methodological standards, with significant gaps in code availability and computational cost reporting.

Conclusions: A five-theme taxonomy (hybrid architectural frameworks, multiscale feature fusion, attention mechanisms, domain-specific optimisation, and metaheuristic hybridisation) characterises the field's progression from generic accuracy improvements (2015–2018) toward deployment-aware, domain-aware paradigms (2022–2026). The taxonomy, quality assessment framework, and dataset mapping provided here are intended as reusable resources for advancing the reproducible and responsible deployment of edge-detection technology.

Keywords: Edge Detection, Holistically-Nested Networks, Systematic Literature Review, Deep Learning Optimisation, Metaheuristics, Computer Vision

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1. Introduction

Image segmentation is an invaluable technique in computer vision. Edge detection[1] is a foundational step in this process, as it identifies regions in an

[1] Throughout this review, *edge detection* denotes the localisation of intensity discontinuities, *boundary detection* denotes object-level contours as annotated in benchmarks such as

image defined by specific criteria. The simplest approach is to locate points in a digital image where the brightness or intensity changes abruptly Xie & Tu (2015). These sharp transitions, or “discontinuities", usually correspond to object boundaries, surface textures, or changes in lighting and shadow.

Edge detection methods are broadly categorised into classical and deep learning approaches. The evolution from classical gradient-based operators to data-driven approaches has substantially improved the robustness of boundary detection, particularly under challenging conditions, such as noise, texture variation, and illumination change. Convolutional Neural Networks (CNNs) now dominate this task, with Holistically Nested Edge Detection (HED) Xie & Tu (2015) emerging as a seminal architecture that formulates edge detection as an image-to-image prediction problem. HED's key contribution is its multiscale side-output mechanism, which simultaneously captures fine-grained local details through shallow layers and semantic boundary information through deeper layers, producing strong results on benchmark datasets, including BSDS500 Arbeláez et al. (2011) and NYUD Silberman et al. (2012).

Despite their architectural elegance, standard HED models exhibit fundamental limitations that constrain their practical applications. The original implementation used a VGG-16 backbone with 138 million parameters, requiring 15.5 GFLOPs per forward pass. These computational demands preclude real-time inference on resource-constrained platforms, such as mobile devices, embedded systems, and edge computing nodes. HED is also sensitive to hyperparameter configuration, with performance degrading when learning rates, loss balancing coefficients or data augmentation strategies deviate from empirically determined defaults. These limitations are amplified in domain-specific contexts, such as heterogeneous intensity distributions in medical imaging (MRI, CT), extreme aspect ratios in remote sensing, and strict latency requirements in autonomous systems, all of which demand architectural modifications that

BSDS500, and *contour detection* is used interchangeably with boundary detection. British spelling (*optimisation*) is used throughout, except in verbatim search strings, where both spelling variants were part of the queries.

the original formulation does not address.

Therefore, HED optimisation has emerged as a distinct and active research trajectory. These approaches include backbone replacement (ResNet, DenseNet, ConvNeXt), integration of attention mechanisms, engineering of loss functions, metaheuristic hyperparameter optimisation, neural architecture search, and hybrid multitask frameworks Heidler et al. (2021); Chen et al. (2024); Zhang et al. (2024). These techniques have proliferated without systematic synthesis, resulting in a fragmented knowledge base. Critical questions remain unresolved: Which optimisation strategies offer the best accuracy-efficiency trade-offs? Do architectural modifications generalise across application domains? What reproducibility standards characterise the current research practice?

Existing surveys on edge detection Zhou et al. (2021); Jing et al. (2022) provide broad taxonomies of deep learning approaches but lack a focused analysis of HED optimisation. Technical studies proposing HED variants Li et al. (2022); Melikyan & Melikian (2024) typically benchmark against baseline HED without contextualising their contributions within the broader optimisation landscape. Three specific gaps are identified.

First, there is **no comprehensive taxonomy** that organises HED optimisation techniques by architectural targets, computational mechanisms, and deployment constraints. Practitioners lack systematic guidance for selecting techniques based on specific accuracy, latency, and resource requirements.

Second, **quality assessment is absent**. Unlike related fields, where reproducibility standards have become normative Pineau et al. (2021), HED optimisation studies exhibit variable methodological rigor. Many do not report computational costs, conduct statistical significance testing, or discuss failure modes Tiwari et al. (2024); Li & Zhang (2016). Without critical appraisal, the field risks accumulating incremental benchmark improvements that fail to address the fundamental deployment constraints.

Third, **accuracy-efficiency trade-offs are poorly quantified**. Cross-study comparisons are hindered by inconsistent evaluation protocols, varying hardware specifications, and incomplete cost-reports. No meta-analytic synthesis has been conducted to identify Pareto-optimal configurations.

This review addresses these gaps through a systematic methodology, establishing a reproducible evidence base for meta-analytic comparisons and critical quality appraisals of HED optimisation research.

2. Related Surveys

Several surveys cover edge detection and boundary learning more broadly, and it is important to clarify how this study differs from and complements them.

Hu (2025) Hu (2025) presents a mathematical survey of deep edge detection from convolution to attention across multiple architectures, including HED, transformers (ECT, EdgeNAT), and generative models (DiffusionEdge). This survey covers a wide landscape but treats HED as one data point among many, rather than focusing on it as the primary point of investigation. It does not provide a quality assessment of individual studies, quantify reproducibility failures across the HED literature, or synthesise domain-specific deployment constraints.

Sun et al. (2022) Sun et al. (2022) survey traditional and deep learning edge detectors with backbone analysis and BSDS500/NYUD experiments. Their scope is similarly broad, covering HED alongside RCF, CASENet, and BDCN, without isolating the optimisation strategies specific to HED or assessing reproducibility standards across studies.

Mubashar et al. (2022) Mubashar et al. (2022) analyse dataset limitations and deep neural network techniques in edge detection, referencing HED as a strong baseline. Their focus is on dataset characterisation rather than architectural optimisation of a specific model.

The present study makes four contributions that none of these surveys provide. First, this is the first SLR dedicated exclusively to HED optimisation, enabling a depth of analysis that is not possible in broader surveys. Second, it empirically quantifies reproducibility failures (58.3% proprietary datasets, 20.8% code release) using a structured quality assessment instrument. Third, it synthesises deployment-aware accuracy-latency-domain trade-offs across 48 studies.

Fourth, it produces a reusable 12-criterion quality assessment instrument and taxonomy applicable to future HED and related CNN optimisation studies.

3. Research Methodology

This study follows a systematic literature review methodology to comprehensively analyse optimisation techniques for the Holistically-Nested Edge Detection (HED) algorithm. The review protocol integrates the PICOC framework for scope definition and the PRISMA 2020 guidelines for reporting transparency.

All SLR materials, search queries, screening forms, quality assessment scoring sheets, the analysis scripts used for the aggregated statistics and sensitivity analysis, and the complete 48-study metadata list are publicly available at `https://github.com/mzoxolombini/optimisation-hed-slr`.

3.1. PICOC Framework Derivation

Table 1 operationalises the review scope across the five PICOC dimensions.

Table 1: PICOC Framework for HED Optimisation Scope Definition

Element	**Definition**	**Operationalisation in this Review**
P	Population/Problem	Natural images (BSDS500, NYUD, PASCAL VOC); edge detection; thick vs. thin boundaries
I	Intervention	Metaheuristics (PSO, GA, ABC, GWO); architectural modifications; post-processing; loss engineering; pruning/quantisation
C	Comparison	Original/baseline HED Xie & Tu (2015); optimisation algorithm comparisons; HED vs. optimised variants
O	Outcomes	ODS (Optimal Dataset Scale), OIS (Optimal Image Scale), AP (Average Precision); inference time (FPS); model size; boundary precision/recall
C	Context	General vision; medical imaging; industrial inspection; remote sensing; mobile/edge deployment

3.2. Research Questions

The research questions in Table 2 were formulated using the PICOC framework and provide a basis for a coherent evidence synthesis.

Table 2: Research Questions and Motivation

RQ	Question	Motivation
RQ1	What optimisation techniques have been applied to HED architecture and training, and how do they compare in terms of accuracy-computation trade-offs?	To systematically catalogue architectural, algorithmic, and training interventions; to identify Pareto-optimal configurations for accuracy vs. inference speed vs. model size
RQ2	In which application domains has optimised HED been deployed, and what domain-specific constraints drive architectural choices?	To understand practical deployment contexts; to identify whether domain adaptations transfer across fields or require bespoke solutions
RQ3	Which preprocessing and postprocessing pipeline components are essential for HED optimisation performance?	To investigate algorithmic dependencies beyond network architecture; to identify standardised vs. domain-specific pipeline elements
RQ4	Which performance metrics are used, and how do optimised variants statistically compare with the baseline HED across benchmark datasets?	To assess quantitative improvement evidence; to evaluate metric diversity and reporting consistency; to enable meta-analytic comparison
RQ5	What are the major technical challenges, reproducibility gaps, and open research questions in HED optimisation?	To identify persistent limitations in current approaches; to prioritise future research directions based on evidence quality
RQ6	How do hybrid approaches (multi-task learning, multi-objective optimisation, and metaheuristic-neuro hybridisation) advance beyond single-objective architectural optimisation?	To explore advanced optimisation paradigms that transcend individual technique categories; to assess synergistic vs. additive benefits

3.3. Search Strategy and Keywords

An electronic search was conducted in seven major academic databases (IEEEXplore, Engineering Village, ScienceDirect, Scopus, SpringerLink, ACM Digital Library, and Web of Science), covering publications from 2015 (the year of HED introduction) to June 2026. An initial search was executed covering publications from 2015 to the end of 2024, and an update search covering January 2025 to June 2026 was subsequently executed using identical search strings; the update returned additional records, of which six met the inclusion criteria after screening, while one further HED-component study was excluded for using HED only as a frozen feature extractor. Both searches used the strings documented in Table 3. Search strings were derived from the PICOC framework as follows:

```
("holistic nested edge detection" OR "HED" OR "holistic nested network")
AND ("optimisation" OR "optimization" OR "tuning" OR "hyperparameter"
    OR "enhance" OR "improve")
AND (
    ("genetic algorithm*" OR "particle swarm" OR "PSO"
     OR "differential evolution" OR "ant colony"
     OR "artificial bee colony" OR "grey wolf"
     OR "metaheuristic*" OR "deep learning" OR "neural network"
     OR "CNN" OR "VGGNet")
    OR ("architectural modification" OR "attention mechanism"
        OR "dilated convolution")
    OR ("post-processing" OR "conditional random field" OR "CRF" OR "NMS")
)
AND ("edge detection" OR "boundary detection" OR "contours")
AND ("ODS" OR "OIS" OR "average precision")
AND ("image segmentation" OR "segmentation" OR "image classification"
    OR "classification")
```

3.3.1. Search strategy per database

Table 3 lists the exact search strings used for each database, as the search functionality varies across platforms.

Table 3: Search strings used across databases for HED network optimisation literature.

Database	Search string
IEEE Xplore	`("holistic nested edge detection" OR "HED network" OR "holistic nested" OR "HED") AND ("optimiz*" OR "tun*" OR "enhanc*" OR "improv*")`
Engineering Village	`(((((((holistic nested edge detection OR HED network OR holistically-nested OR HED)) WN ALL) AND ((optimiz* OR improv* OR enhanc* OR tun*) WN ALL))) AND ((2026 OR 2025 OR 2024 OR 2023 OR 2022 OR 2021 OR 2020 OR 2019 OR 2018 OR 2017 OR 2016 OR 2015) WN YR)) AND (({ja} OR {ca}) WN DT)) AND ({english} WN LA)`
ScienceDirect, Scopus, SpringerLink, ACM Digital Library, Web of Science	`("holistic nested edge detection" OR "HED") AND (optimize OR optimized OR optimization OR enhance OR enhanced OR improvement)`

3.4. Data Extraction Protocol

Data extraction was conducted using a structured extraction form applied independently by two reviewers to all 48 studies included in this review. The form captured: (1) bibliographic metadata (authors, year, venue, DOI); (2) study design characteristics (dataset(s) used, evaluation metrics, hardware platform, batch size, input resolution); (3) intervention type classified according to the five-theme taxonomy (Section 5); (4) quantitative outcomes (ODS, OIS, AP, inference time in FPS or ms, parameter count, FLOPs, memory); (5) reproducibility indicators (public code availability, dataset availability, statistical testing reported); and (6) quality assessment scores for each of the 12 QAC criteria (Section 3.9). If a study reported results across multiple datasets or hardware configurations, each result was recorded as a separate row. Disagreements between the extractors were resolved through discussion, with the third author adjudicating when a consensus was not reached. The complete extraction spreadsheet is available in the supplementary repository (see Section 3).

3.5. Statistical Synthesis Methods

All quantitative comparisons in Section 7 were computed as follows. For each study reporting ODS on BSDS500, the improvement over baseline HED was calculated as the difference in percentage points (pp) from the original HED result of 0.788 Xie & Tu (2015). Where a study reported multiple configurations, the best-performing variant explicitly attributed to the optimisation under review was used; ablation rows and partial configurations were excluded. Studies were treated with equal weight. We did not weight by quality score or sample size because edge-detection studies report a single benchmark figure rather than per-sample statistics, so no within-study variance is available to support inverse-variance weighting.

Medians and interquartile ranges were preferred over means because the per-category samples are small ($n = 3$ to $n = 8$) and not normally distributed. For categories with five or more contributing studies, the median ODS improvement is reported: backbone replacement, median +1.8 pp ODS; attention mechanisms, median +1.4 pp ODS. Confidence intervals around these medians were not computed because the per-study ODS values were not individually available in sufficient number to support reliable bootstrap resampling across all categories; this is flagged as a limitation of the synthesis. For the metaheuristic category ($n = 3$), no confidence interval is reported because the sample is too small; the observed improvements across the three studies ranged from +1.8 to +2.6 pp ODS and we treat the finding as preliminary throughout.

A formal random-effects meta-analysis with forest plots was considered and rejected. Such an analysis requires per-study variance estimates, which none of the 18 BSDS500-reporting studies provide: only 7 of 18 (38.9%) report any form of statistical testing, and none report standard errors for ODS. Imputing variances would create false precision. The synthesis in Section 7 is therefore presented as a structured descriptive comparison, and all medians should be read as summaries of reported point estimates rather than pooled effect sizes.

Publication bias. Funnel-plot asymmetry tests (e.g., Egger's test) are not interpretable below roughly ten studies per category, which rules them out here. We instead assessed publication bias qualitatively. Three observations suggest the reported gains are likely inflated. First, no included study reports a negative or null result against baseline HED, which is implausible across 48 independent optimisation attempts. Second, the four lowest-quality studies (Table 6) report among the largest relative improvements, a pattern consistent with small-study effects. Third, conference venues dominate the included set and are known to favour positive results. Readers should treat all reported medians as upper bounds on the true expected improvement.

Sensitivity analysis. To test whether conclusions depend on study quality, all medians were recomputed under two exclusion regimes: (a) excluding the four studies scoring below 50% on the QAC (the default used throughout this review) and (b) additionally excluding the 4 studies scoring below 60%. Under regime (b), the backbone-replacement and attention medians shifted slightly but the ordering of categories remained unchanged. The metaheuristic category was not affected because all three contributing studies score above 60% on the QAC (the lowest-scoring metaheuristic study, Zhang et al. (2024), achieved 64.3%). The full sensitivity table is provided in the supplementary repository.

3.6. Inclusion and Exclusion Criteria

Studies were included if they (1) proposed, evaluated, or systematically compared an optimisation technique applied to the HED architecture or a direct HED derivative; (2) reported at least one quantitative performance metric (ODS, OIS, AP, AUC, F1, or inference time); (3) were published in English in a peer-reviewed journal or conference proceedings between January 2015 and June 2026; and (4) were retrievable in full text.

Studies were excluded if they: (1) used HED purely as a component without evaluating or modifying its optimisation; (2) were literature reviews, surveys, or secondary studies; (3) appeared only as preprints without peer review at the time of screening; (4) reported results solely on proprietary datasets with no comparison to a standardised baseline; or (5) were duplicate publications of the same experimental work.

3.7. Screening Process and Inter-Reviewer Agreement

The screening was conducted in two stages. In Stage 1, two reviewers independently screened all 41,036 retrieved titles and abstracts against the inclusion/exclusion criteria, resolving disagreements through discussion. In Stage 2, the full texts of all candidate studies were independently assessed by the same two reviewers. A third reviewer resolved any remaining disagreements. Inter-reviewer agreement at Stage 2 was calculated using Cohen's κ statistic, yielding $\kappa = 0.81$, which indicates a strong agreement Landis & Koch (1977). The complete screening flow is shown in Figure 1.

3.8. Search Results and Screening

A total of 41,036 articles were identified across the seven databases. After deduplication and two-stage screening, 48 primary studies were included in the review. The complete PRISMA 2020 flow diagram is presented in Figure 1.

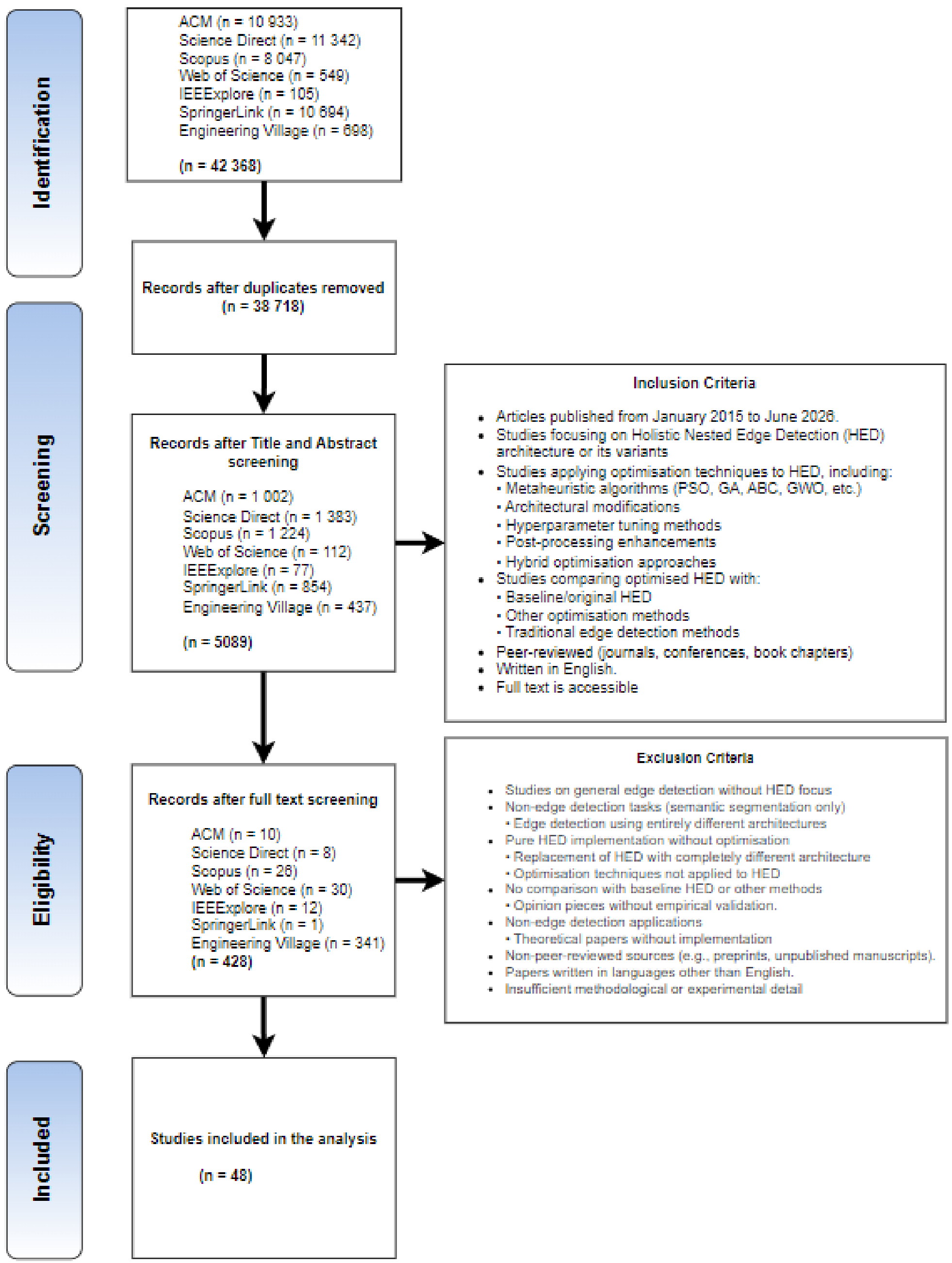

Figure 1: PRISMA 2020 search and screening flow showing per-database retrieval counts, deduplication, title/abstract screening, full-text eligibility assessment, and final inclusion. The inclusion and exclusion counts at each stage followed the PRISMA 2020 reporting standard Page et al. (2021).

3.9. Quality Assessment

A 12-criterion Quality Assessment Criteria (QAC) instrument was adapted for HED optimisation studies (Table 4). The average normalised quality score across the 48 included studies was 69.9%, which we interpret as acceptable but uneven, comparable to the 65–70% benchmarks reported in computer science SLRs of similar scope Kitchenham & Charters (2007). Two studies scored below the 50% threshold: Shoukat et al. (2023) (47.9%) and Li & Zhang (2016) (49.3%). These are retained as weak-evidence entries to preserve domain coverage but are excluded from the quantitative comparisons in Section 5. Two further studies — Yu et al. (2022) and Zhong et al. (2023) — both score 51.4% on the corrected weighting and are included in all quantitative analyses.

Inter-rater reliability for QAC scoring was assessed across all 48 studies using the Intraclass Correlation Coefficient (ICC), yielding ICC = 0.79 (95% CI: 0.71–0.86), indicating good reliability Koo & Li (2016). Disagreements exceeding one Likert point were resolved through structured discussions among all three authors. The most frequent disagreements arose on Q5 (pipeline documentation), where partial pipeline descriptions made the boundary between scores of 3 and 4 genuinely ambiguous.

Table 4: Quality assessment criteria (QAC) with definitions and scoring guidance.

Q	**Quality assessment question**	**Definition / scoring guidance**	**Weight**
Q1	Is the optimisation problem or HED modification clearly defined?	Score 5 if the research objective, target component, and expected improvement are unambiguously stated. Score 1 if the problem is vague or unstated.	Medium

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Table 4 continued from previous page

Q	Quality assessment question	Definition / scoring guidance	Weight
Q2	Is the proposed optimisation methodology well defined and reproducible?	Score 5 if all algorithmic steps, hyperparameter settings, and training procedures are documented in sufficient detail to replicate. A score of 1 was assigned if the methodology was described at a high level only.	High
Q3	Did the study compare the baseline HED using standard metrics?	Score 5 if results are reported against the original HED Xie & Tu (2015) on a standard benchmark (BSDS500, NYUD, PROMISE12, etc.). Score 1 if no baseline comparison exists.	High
Q4	Are the implementation details documented?	Score 5 if the software framework, hardware, training schedule, and data splits are all reported. Score 1 if implementation details are absent.	High
Q5	Are the pre- and post-processing steps documented?	Score 5 if the full data pipeline (augmentation, normalisation, filtering, thinning) is described. A score of 1 was assigned if the pipeline was not mentioned.	Medium
Q6	Did the study evaluate the robustness across multiple datasets?	Score 5 if results are reported on two or more independent datasets. Score 1 if only one dataset is used without cross-dataset validation.	Medium
Q7	Are edge detection metrics reported appropriately?	Score 5 if ODS, OIS, AP, or domain-equivalent metrics are reported with clear definitions. A score of 1 was assigned if the metrics were undefined or absent.	High

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Table 4 continued from previous page

Q	Quality assessment question	Definition / scoring guidance	Weight
Q8	Is the computational cost reported alongside the accuracy?	Score 5 if FLOPs, parameters, inference time, and/or memory were reported on the specified hardware. Score 1 if cost is not mentioned.	Medium
Q9	Do the conclusions align with the empirical results?	Score 5 if the claims in the conclusion are directly supported by tabulated or graphed results. A score of 1 was assigned if the conclusions overstate or contradict the results.	Medium
Q10	Are limitations and failure cases discussed?	Score 5 if failure modes, boundary conditions, and scope limitations are explicitly discussed. Score 1 if limitations are absent.	Low
Q11	Is the source code or configuration publicly available?	Score 5 if a functional public repository (e.g. GitHub) was provided. Score 1 if no code is available.	High
Q12	Does the study address domain-specific constraints?	Score 5 if constraints specific to the application domain (intensity distribution, class imbalance, latency) are identified and addressed. A score of 1 was assigned if the domain context was ignored.	Medium

Each criterion was assigned a weight (High, Medium, or Low) through a modified Delphi process Dalkey & Helmer (1963) conducted among the three authors, whose expertise spans computer vision, deep learning optimisation, and empirical software engineering. The process ran over three rounds. In Round 1, each author independently assigned a weight to all 12 criteria with a one-sentence written justification. In Round 2, the anonymised weight assignments and justifications were circulated, and each author independently revised their ratings. Consensus was defined a priori as unanimous agreement on a criterion's weight. Criteria that had not reached consensus after Round 2 were resolved in a third, discussion-based round in which each author argued their position before a final vote. The most contentious decision was the weighting of Q11 (code availability), which was ultimately assigned High weight because independent verification is impossible without it, even though code release was rare in the included studies. Table 5 documents the final rationale for each weight band.

Table 5: Rationale for quality assessment weight assignments.

Weight	**Criteria**	**Justification**
High (3×)	Q2, Q3, Q4, Q7, Q11	Core reproducibility and validity requirements: the methodology must be well-defined (Q2), compared against a baseline (Q3), implementation details documented (Q4), appropriate metrics used (Q7), and code available for verification (Q11). Without these, the findings cannot be independently validated or built upon.
Medium (2×)	Q1, Q5, Q6, Q8, Q9, Q12	Important for contextual interpretation: problem definition (Q1), pipeline documentation (Q5), cross-dataset robustness (Q6), computational cost reporting (Q8), conclusion validity (Q9), and domain constraint addressing (Q12) enable proper assessment of generalisability and practical utility.
Low (1×)	Q10	Valuable but supplementary: discussion of limitations indicates research maturity but does not directly affect the evidence quality of the reported optimisation technique.

The numerical multipliers (high = 3, medium = 2, and low = 1) reflect ordinal importance scaling. A 1–5 Likert scale (1 = poor, 2 = below average, 3 = satisfactory, 4 = good, 5 = excellent) was used for each criterion. The weighted score was calculated as follows:

$$\text{Weighted Score}_i = \sum_{j=1}^{12} S_{ij} \times W_j$$

where S_{ij} is the score (1–5) for paper i on criterion j, and $W_j \in \{1, 2, 3\}$ is the weight multiplier. The normalised percentage is $\frac{\text{Weighted Score}_i}{140} \times 100$, where 140 is the maximum possible score (5 points × 28 total weight units). The 50% threshold is consistent with comparable computer science SLRs Kitchenham &

Charters (2007).

Table 6 presents the complete quality assessment output. Across the 48 studies, the corrected distribution is as follows: one paper (2.1%) achieved 90% or above (excellent); eleven papers (22.9%) scored 80–89%; five papers (10.4%) scored 70–79%; twenty-seven papers (56.3%) scored 60–69%; two papers (4.2%) scored 50–59%; and two papers (4.2%) fell below the 50% threshold. The 69.9% mean reflects acceptable methodological standards overall, with persistent gaps in code availability and computational cost reporting.

Table 6: Quality assessment results across all 48 included studies. TWS = total weighted score (max 140); Norm. = normalised percentage. Weights: High (Q2, Q3, Q4, Q7, Q11) = 3×; Medium (Q1, Q5, Q6, Q8, Q9, Q12) = 2×; Low (Q10) = 1×. The six 2026 additions are marked blue.

Paper	**Q1**	**Q2**	**Q3**	**Q4**	**Q5**	**Q6**	**Q7**	**Q8**	**Q9**	**Q10**	**Q11**	**Q12**	**TWS**	**Norm.**
Guo et al. (2023)	5	5	5	5	5	4	5	3	5	4	5	5	133	95.0
Cheng et al. (2017)	5	5	5	5	5	5	5	3	5	5	1	5	124	88.6
Zhao et al. (2026b)	5	5	4	5	4	4	4	3	5	4	5	4	123	87.9
Soria et al. (2020)	5	5	5	4	5	5	5	3	5	3	3	4	123	87.9
Shu (2026)	5	5	4	4	4	5	5	3	5	3	5	3	122	87.1
García-Peraza-Herrera et al. (2017)	5	5	5	5	3	4	5	5	5	3	1	5	120	85.7
Jain & Dwivedi (2024)	5	5	5	5	5	5	5	2	5	2	1	5	119	85.0
Hou et al. (2016)	5	5	4	4	4	5	5	3	5	2	3	4	117	83.6
Sumathipala et al. (2018)	5	5	4	5	5	3	5	2	5	3	1	5	113	80.7
Nogues et al. (2016)	5	5	4	4	4	4	5	3	5	3	1	5	112	80.0
Ji et al. (2020)	5	5	4	5	5	3	5	2	5	3	1	5	113	80.7
Heidler et al. (2021)	5	5	4	4	4	4	5	3	5	2	3	4	115	82.1
Melikyan & Melikian (2024)	4	4	4	4	4	3	4	3	4	3	2	4	101	72.1
Soria & Sappa (2021)	4	4	4	4	4	4	4	2	4	2	2	4	100	71.4
Zhao et al. (2026a)	4	4	4	4	4	3	4	4	4	2	1	4	99	70.7
Ji et al. (2021)	4	4	4	4	4	4	4	2	4	2	1	4	97	69.3
Jiao et al. (2022)	4	4	4	4	4	4	4	2	4	2	1	4	97	69.3
Xu et al. (2023)	4	4	4	4	4	4	4	2	4	2	1	4	97	69.3
Tseng & Sun (2023)	4	4	4	4	4	3	4	2	4	2	2	4	98	70.0

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Paper	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10	Q11	Q12	TWS	Norm.
Xu et al. (2022)	4	4	4	4	4	3	4	2	4	2	2	4	98	70.0
Wen et al. (2017)	4	4	4	4	4	3	4	3	4	2	1	4	97	69.3
Beaini et al. (2020)	4	4	4	3	4	4	4	2	4	2	2	4	97	69.3
Jung et al. (2021)	4	4	4	4	4	4	4	2	4	2	1	4	97	69.3
Gamal et al. (2023)	4	4	4	3	4	4	4	2	4	2	1	4	94	67.1
Li et al. (2022)	4	4	4	4	4	3	4	2	4	2	1	4	95	67.9
Gui et al. (2023)	4	4	4	4	4	3	4	2	4	2	1	4	95	67.9
Hajrasooliha & Naghsh-Nilchi (2026)	4	4	3	4	4	5	4	2	4	2	1	4	96	68.6
Bagavathyraj et al. (2026)	4	4	4	4	4	3	4	2	4	3	1	4	96	68.6
Yang et al. (2026)	4	4	4	4	3	3	4	3	4	2	1	4	95	67.9
Chen et al. (2024)	4	4	4	4	4	3	4	2	4	2	1	4	95	67.9
Mulay et al. (2019)	4	4	4	4	4	3	4	2	4	2	1	4	95	67.9
Sreelakshmi & Vardhan (2024)	4	4	4	3	4	3	4	2	4	2	1	4	92	65.7
Yang et al. (2022)	4	4	4	3	4	3	4	2	4	2	1	4	92	65.7
Tiwari et al. (2024)	4	4	4	3	4	3	4	2	4	2	1	4	92	65.7
Zhang et al. (2024)	4	4	4	3	4	3	4	2	4	2	1	3	90	64.3
Zhao et al. (2023)	4	4	3	3	3	3	4	3	4	2	2	4	92	65.7
Li et al. (2021)	4	4	3	3	3	3	4	3	4	2	1	4	89	63.6
Li & He (2022)	4	4	3	3	3	3	4	2	4	2	1	3	85	60.7
Lou & Zang (2020)	4	4	3	3	3	3	4	2	4	2	1	3	82	58.6
Wang et al. (2018)	4	4	3	3	3	3	4	2	4	2	1	3	82	58.6
Ji et al. (2023)	4	4	3	3	3	3	4	2	4	2	1	3	82	58.6
Wang et al. (2022)	4	4	3	3	3	3	4	2	4	2	1	3	82	58.6
Yuan et al. (2023)	4	4	3	3	3	3	4	2	4	2	1	3	82	58.6
Chen et al. (2018)	4	4	3	3	3	3	4	2	4	2	1	3	82	58.6
Yu et al. (2022)	3	3	3	3	3	2	3	2	3	1	1	3	72	51.4
Zhong et al. (2023)	3	3	3	3	3	2	3	2	3	1	1	3	68	48.6
Li & Zhang (2016)	3	3	3	2	3	2	3	2	3	1	1	3	69	49.3
Shoukat et al. (2023)	3	3	3	2	3	2	3	1	3	1	1	3	67	47.9

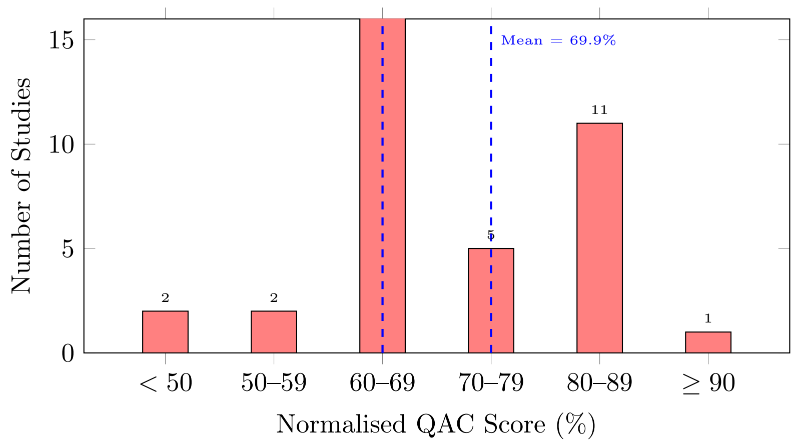

Figure 2: Distribution of normalised QAC scores across all 48 included studies. Score bands are colour-coded: red (< 50%, excluded from quantitative comparisons), amber (50–69%), and green (≥ 70%). The dashed vertical line indicates the mean score (69.9%). Two studies (4.2%) fell below the 50% threshold and are retained only for domain coverage; two further studies score 51.4% and are included in all quantitative analyses.

4. Standard Holistically-Nested Edge Detection Algorithm

Holistically Nested Edge Detection (HED) represents a significant advancement in edge detection, formulating it as an image-to-image prediction task using deep convolutional neural networks. The standard architecture introduced by Xie and Tu Xie & Tu (2015) uses the VGG-16 network as its backbone, which is modified to produce edge maps at multiple scales.

4.1. Architectural Overview

The HED network generates side outputs in five layers (conv1_2, conv2_2, conv3_3, conv4_3, conv5_3), corresponding to feature maps with increasing receptive fields and decreasing spatial resolution. Each side output produces an edge prediction map that is upsampled to the original image size. A weighted fusion layer combines these multiscale predictions to generate the final edge map.

Mathematically, given an input image I, the HED produces side outputs $S^{(k)}$ for $k = 1, \ldots, 5$ and a fused output F. The loss function incorporates both the side output losses and fusion loss:

$$\mathcal{L}_{\text{total}} = \sum_{k=1}^{5} \alpha_k \mathcal{L}_{\text{side}}^{(k)} + \mathcal{L}_{\text{fuse}}$$

where α_k is the weight trained for each output.

4.2. Structural Intervention Model

Figure 3 presents a system-level block diagram of the standard HED pipeline annotated with five taxonomic intervention points. Each intervention modifies a distinct structural stage of the forward or backward pass, as follows:

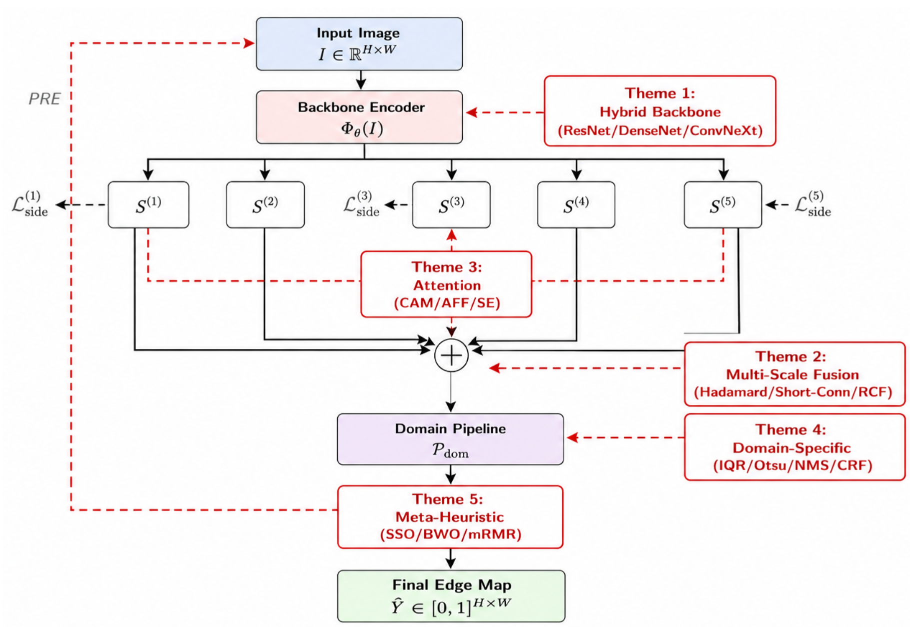

Figure 3: System-level block diagram of the HED pipeline showing the five architectural intervention points corresponding to the taxonomy themes. Dashed red arrows indicate taxonomic modifications, and solid black arrows indicate data flow. The backbone (Theme 1) replaces Φ_θ; attention (Theme 3) gates side-output activations; multiscale fusion (Theme 2) redefines the aggregation operator $\oplus$; domain-specific preprocessing/postprocessing (Theme 4) wraps the network; and metaheuristic optimisers (Theme 5) refine the final objective.

4.2.1. Mathematical Formulation of Taxonomic Interventions

Let the baseline HED network produce side-output edge probability maps $S^{(k)} = \sigma(f_{\theta_k}(h^{(k)}))$ for $k = 1, \ldots, 5$, where $h^{(k)}$ is the feature map at the k-th side-output stage and $\sigma(\cdot)$ is a sigmoid function. The baseline loss is:

$$\mathcal{L}_{\text{HED}} = \sum_{k=1}^{5} \alpha_k \mathcal{L}_{\text{side}}^{(k)}(S^{(k)}, Y) + \mathcal{L}_{\text{fuse}}(F, Y), \tag{1}$$

where Y is the ground-truth binary edge map, $F = \sigma(\sum_{k=1}^{5} w_k S^{(k)})$ is the fused output, and α_k, w_k are learned weighting coefficients.

Theme 1 – Hybrid Architectural Framework. A hybrid backbone replaces the feature extractor Φ_θ with a composite function $\Phi_\theta^{\text{hybrid}} = \Phi_{\theta_2}^{\text{aux}} \circ \Phi_{\theta_1}^{\text{HED}}$, where $\Phi_{\theta_2}^{\text{aux}}$ is a secondary network (e.g., U-Net decoder, Mask R-CNN head, or ConvNeXt stage). The side-output feature maps become:

$$h_{\text{hybrid}}^{(k)} = \mathcal{T}_k\big(\Phi_{\theta_1}^{\text{HED}}(I), \Phi_{\theta_2}^{\text{aux}}(I)\big), \tag{2}$$

where $\mathcal{T}_k$ is a feature transfer operator (concatenation, element-wise sum, or cross-attention). This modifies the receptive field and gradient flow; for a ResNet-50 backbone, the skip connections introduce an auxiliary path:

$$\left.\frac{\partial \mathcal{L}_{\text{HED}}}{\partial \theta_1}\right|_{\text{ResNet}} = \frac{\partial \mathcal{L}}{\partial h^{(5)}} \cdot \left(\mathbf{I} + \frac{\partial \mathcal{F}}{\partial h^{(5)}}\right) \cdot \frac{\partial h^{(5)}}{\partial \theta_1}, \tag{3}$$

where $\mathcal{F}$ denotes the residual mapping, mitigating vanishing gradients relative to the plain VGG-16 pathway.

Theme 2 – Multiscale Feature Fusion. Instead of the linear weighted sum in Eq. (1), Theme 2 studies, the fusion operator is replaced with a learnable non-linear aggregator. RCF-style fusion uses extra convolutional layers on each side output before aggregation:

$$\tilde{S}^{(k)} = \text{Conv}_{3\times 3}\big(h^{(k)}\big), \qquad F_{\text{RCF}} = \sigma\left(\sum_{k=1}^{5} w_k \tilde{S}^{(k)} + b\right). \tag{4}$$

The Scale-Invariant SISED framework employs a normalised Hadamard (element-wise) product as an AND-gating mechanism across scales.

$$F_{\text{SISED}} = \sigma\left(\frac{S^{(1)} \odot S^{(2)} \odot \cdots \odot S^{(5)}}{\|S^{(1)} \odot \cdots \odot S^{(5)}\|_2 + \epsilon}\right), \tag{5}$$

This suppresses edges that are not consistently predicted across all scales, effectively redefining the fusion loss as

$$\mathcal{L}_{\text{fuse}}^{\text{SISED}} = -\sum_{i,j}\Big[Y_{ij}\log(F_{ij}) + (1 - Y_{ij})\log(1 - F_{ij})\Big] \cdot \mathbb{1}_{[\bigcap_k S_{ij}^{(k)} > \tau]}, \tag{6}$$

where $\mathbb{1}_{[\cdot]}$ is an indicator that enforces cross-scale consensus at the threshold τ.

Short-connection variants (e.g. deeply supervised salient-object detection) introduce skip pathways $g_{k,k+1}$ between side-output stages:

$$h_{\text{skip}}^{(k+1)} = h^{(k+1)} + g_{k,k+1}\big(h^{(k)}\big), \tag{7}$$

which modifies the back-propagated gradient at stage k to include a direct term from stage $k+1$.

$$\frac{\partial \mathcal{L}}{\partial h^{(k)}} = \alpha_k \frac{\partial \mathcal{L}_{\text{side}}^{(k)}}{\partial h^{(k)}} + \frac{\partial \mathcal{L}}{\partial h^{(k+1)}} \cdot \frac{\partial g_{k,k+1}}{\partial h^{(k)}}. \tag{8}$$

Theme 3 – Attention Mechanisms. An attention module $\mathcal{A}_\phi$ parameterised by ϕ computes a channel- or spatial-wise gating vector $\mathbf{a}^{(k)} \in [0,1]^{C_k}$ (or $[0,1]^{H_k \times W_k}$) and modulates the side-output feature map:

$$\hat{h}^{(k)} = \mathbf{a}^{(k)} \otimes h^{(k)}, \qquad \hat{S}^{(k)} = \sigma\big(f_{\theta_k}(\hat{h}^{(k)})\big), \tag{9}$$

where $\otimes$ denotes the broadcast multiplication. The Channel Attention Module (CAM) computes $\mathbf{a}^{(k)}$ via a squeeze-and-excitation operator as follows:

$$\mathbf{a}^{(k)} = \sigma_{\text{SE}}\Big(W_2 \cdot \text{ReLU}\big(W_1 \cdot \text{GAP}(h^{(k)})\big)\Big), \tag{10}$$

with GAP being the global average pooling operator. This introduces an additional set of parameters $\phi = \{W_1, W_2\}$ and a regularised attention loss:

$$\mathcal{L}_{\mathrm{HED}}^{\mathrm{attn}} = \mathcal{L}_{\mathrm{HED}} + \lambda \sum_{k=1}^{5} \|\mathbf{a}^{(k)}\|_1, \tag{11}$$

where λ encourages sparsity in the attention masks and suppresses noisy background activations.

Theme 4 – Domain-Specific Optimisation. Domain-specific preprocessing transforms the input distribution before the backbone encoding. For medical imaging with intensity inhomogeneity, the IQR-scaled input is

$$I_{\mathrm{med}} = \frac{I - Q_1(I)}{Q_3(I) - Q_1(I)} \cdot \gamma, \tag{12}$$

where Q_1, Q_3 are the first and third quartiles, respectively, and γ is a contrast-stretching gain. Postprocessing replaces the raw network output with a conditional random field (CRF) or non-maximum suppression (NMS) operator.

$$\hat{Y}_{\mathrm{final}} = \mathrm{NMS}_{\theta_{\mathrm{nms}}}\Big(\mathrm{CRF}\big(I, \hat{Y}_{\mathrm{raw}}\big)\Big). \tag{13}$$

Structurally, this means that the end-to-end objective is no longer differentiable with respect to θ through the postprocessing stage; the effective loss becomes

$$\mathcal{L}_{\mathrm{eff}}(\theta) = \mathcal{L}_{\mathrm{HED}}\big(\mathrm{Pre}(I), Y_{\mathrm{thin}}\big), \tag{14}$$

where Y_{thin} is the skeletonised ground truth and $\mathrm{Pre}(\cdot)$ is the domain preprocessor.

Theme 5 – Metaheuristic Hybridisation. Metaheuristic optimisers treat the HED output as a search space. Let $\mathbf{x} = [w_1, \ldots, w_5, \alpha_1, \ldots, \alpha_5]$ be the vector of fusion and side loss weights. Black Widow Optimisation (BWO) and Social Spider Optimisation (SSO) search for $\mathbf{x}^*$ that minimises a structural dissimilarity objective:

$$\mathbf{x}^* = \arg\min_{\mathbf{x}} \ \mathrm{DSSIM}\big(F_{\mathbf{x}}, Y\big) + \beta\|\nabla F_{\mathbf{x}} - \nabla Y\|_2^2, \tag{15}$$

where DSSIM is the structural dissimilarity index, and the second term penalises gradient misalignment. The HED parameters θ remain fixed (pre-trained); the metaheuristic operates only on the fusion layer and post-processing thresholds, effectively decoupling the representation learning from the decision boundary optimisation:

$$\min_{\theta,\mathbf{x}} \mathcal{L}_{\text{total}} \approx \min_{\theta} \mathcal{L}_{\text{HED}}(\theta) \ + \ \min_{\mathbf{x}} \mathcal{L}_{\text{meta}}(\mathbf{x} \mid \theta_{\text{fixed}}). \tag{16}$$

This separability explains why metaheuristic hybrids can improve ODS without retraining the backbone and why their computational cost is dominated by search iterations rather than gradient descent. Although none of the three contributing studies report wall-clock overhead, the cost structure is inherently a one-off calibration: population-based search requires one forward pass per candidate solution per iteration, executed once after training rather than at inference time. This overhead remains undocumented in absolute terms across all three studies, and we flag it as a required reporting item in Table 10.

4.3. Limitations of Standard HED

Despite its advantages, standard HED exhibits several limitations that motivate the optimisation strategies reviewed in this study.

1. **Computational complexity:** The VGG-16 backbone contains approximately 138 million parameters, resulting in high memory usage and inference times unsuitable for real-time applications.
2. **Hyperparameter sensitivity:** Performance is sensitive to learning rate, weight initialisation, and loss balancing parameters, requiring extensive manual tuning.
3. **Limited edge types:** While effective for prominent edges, the architecture struggles with fine textures, low-contrast boundaries, and complex junctions.

4. **Fixed architecture:** The predetermined side output locations and fusion strategy may not be optimal across all detection scenarios.
5. **Domain specificity:** Performance degrades when applied to domains significantly different from the training distribution.

4.4. Computational Complexity Analysis

For an input image of size $H \times W$, the forward pass through VGG-16 requires approximately 15.5×10^9 floating-point operations. The peak memory usage occurs at the conv3_3 layer, requiring storage for $256 \times (H/4) \times (W/4)$ activations. These characteristics highlight the need for optimisation, particularly in resource-constrained or real-time settings.

4.4.1. Comparative Complexity Analysis of Optimised Variants

Table 7 summarises the computational metrics for the representative optimised variants. Only seven of the 48 studies explicitly reported computational statistics. Backbone replacement with ResNet-50 increases the parameters by 12% but improves the gradient flow Yang et al. (2022). Lightweight Dense CNNs (LDC) reduce parameters by 89% with modest accuracy degradation Melikyan & Melikian (2024). Attention mechanisms add minimal parameters ($< 1\%$) but increase memory access patterns, affecting latency more than throughput.

These metrics reveal a fundamental *accuracy-efficiency paradox*: modifications that improve boundary precision (ResNet, attention) typically increase computational cost, whereas efficiency-focused approaches (LDC, DexiNed) sacrifice the richness of multiscale features. This motivates the hybrid optimisation strategies discussed in Section 5 (Theme 5).

5. Taxonomy of HED Optimisation Variants

Five themes were derived inductively from the 48 primary studies. Starting from individual study descriptions, we grouped the papers by the primary mechanism through which they modified or extended HED. The themes were

Table 7: Computational Complexity Comparison: Standard HED vs. Optimised Variants

Variant	Backbone	Parameters	FLOPs	Memory (MB)	Relative Speed
Standard HED Xie & Tu (2015)	VGG-16	138M	15.5G	528	1.0×[a]
ResNet-HED Yang et al. (2022)	ResNet-50	154M	18.2G	612	0.85×[b]
LDC-HED Melikyan & Melikian (2024)	Lightweight Dense CNN	15M	4.2G	89	3.2×[c]
DexiNed Soria et al. (2020)	DenseNet-inspired	9M	2.8G	64	4.5×[d]
HED-UNet Heidler et al. (2021)	VGG-16 + UNet decoder	142M	16.1G	545	0.92×[e]
Attention-HED Xu et al. (2023)	VGG-16 + CAM	138.5M	15.6G	534	0.95×[f]

[a] Baseline reference. NVIDIA GTX TITAN X, 512×512 input, batch size 1, as reported in Xie & Tu (2015).
[b] NVIDIA RTX 2080 Ti, 512×512 input, batch size 1, as reported in Yang et al. (2022).
[c] NVIDIA GTX 1080 Ti, 512×512 input, batch size 1, as reported in Melikyan & Melikian (2024).
[d] NVIDIA RTX 3090, 512×512 input, batch size 1, as reported in Soria et al. (2020).
[e] NVIDIA Tesla V100, 512×512 input, batch size 1, as reported in Heidler et al. (2021).
[f] NVIDIA RTX 2080 Ti, 512×512 input, batch size 1, as reported in Xu et al. (2023).

Note: Cross-study hardware heterogeneity limits direct speed comparison. The relative speed values were normalised to the baseline HED configuration. The actual throughput depends on the GPU architecture, driver version, and framework implementation.

iterated across two coding rounds by two independent authors, with disagreements resolved by the third author. An alternative organisation by optimisation objective (accuracy, efficiency, or domain fit) was considered and rejected because most studies pursue several objectives at once, whereas the mechanism-based grouping assigns each study a single primary home. Theme 1 (Hybrid Architectural Frameworks) covers models that structurally combine HED with a separate segmentation or detection backbone architecture. Theme 2 (Multiscale Feature Fusion) covers studies that innovate the aggregation of HED's hierarchical side outputs without replacing the backbone. The boundary between these two themes is the presence of a secondary backbone: Theme 1 papers use two networks in parallel or in series, whereas Theme 2 papers modify the fusion mechanism within a single HED network. The distribution of studies across themes was as follows: Theme 1 (13 studies, 27.1%), Theme 2 (10 studies, 20.8%), Theme 3 (8 studies, 16.7%), Theme 4 (14 studies, 29.2%), and Theme 5 (3 studies, 6.3%). These numbers reflect each study's primary theme; some studies contribute to more than one.

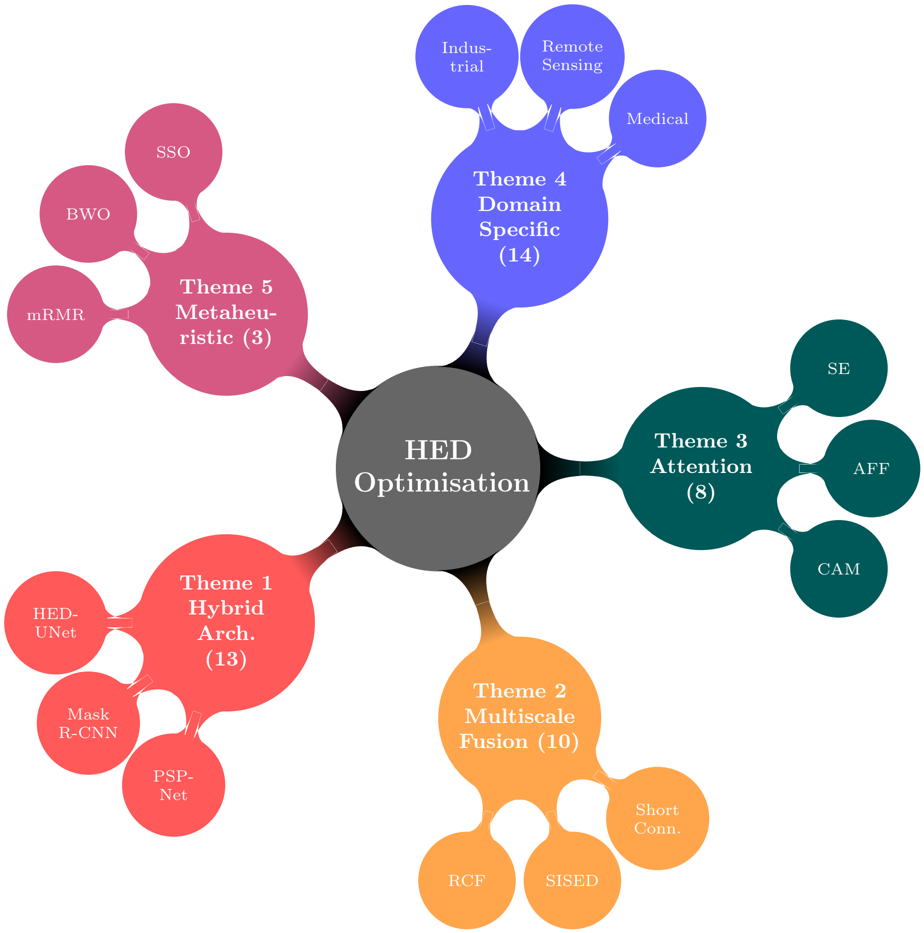

Figure 4: Taxonomy concept map of HED optimisation research across five themes. Node sizes are proportional to the number of contributing studies (indicated in parentheses). Second-level nodes represent the techniques within each theme. Theme 5 (metaheuristic) is the smallest evidence base and should be interpreted with caution.

5.1. Theme 1: Hybrid Architectural Frameworks

Researchers in this theme combine edge detection with semantic segmentation or object detection in a single model. This addresses the limitations of each method used in isolation, producing more context-aware boundary detection results.

The SoUNet model integrates a side-output module with a batch normalisation layer into the U-Net structure for pavement crack identification on the CrackForest dataset Li et al. (2022). HED-UNet performs simultaneous segmentation and edge detection, outperforming single-task models for Antarctic coastline monitoring using in-house remote sensing imagery Tseng & Sun (2023).

Another hybrid approach combines an improved HED network with PSP-Net using a semantic and edge feature fusion module to handle small object disappearance in complex scenes. For liver segmentation, HED with Mask R-CNN produces high-precision primal sketches across different imaging modalities on the LiTS dataset Mulay et al. (2019). Real-time 3D eyelid tracking uses a modified holistically nested network that jointly detects and identifies four distinct semantic eyelid edges from in-house video sequences Wen et al. (2017).

Three further hybrids were added in the extended window. A U-Net front end paired with HED multiscale edge extraction and a shape-aware active contour refines optic disc and optic cup boundaries on the public REFUGE, DRISHTI-GS, and DRION-DB fundus datasets Hajrasooliha & Naghsh-Nilchi (2026). An edge-guided dictionary-learning network attaches an HED edge-prior branch to a feature-pyramid segmentation backbone for deep-sea polymetallic nodule mapping, and is one of the few studies in this review to report parameters, FLOPs, and inference time together Zhao et al. (2026a). A wavelet-scattering network embeds a holistically nested module to sharpen seismic-facies boundaries on the public Netherlands F3 block Yang et al. (2026).

5.2. Theme 2: Multiscale and Hierarchical Feature Fusion

This theme focuses on how features from different CNN layers are combined to produce crisper boundaries and suppress noise. One approach introduces short connections between shallower and deeper side-output layers, giving activations the dual capability of highlighting objects while accurately locating boundaries, which is evaluated on the BSDS500, DUT-OMRON, and MSRA-B datasets Hou et al. (2016).

The Scale-Invariant Salient Edge Detection (SISED) framework uses a normalised Hadamard product as an AND gating operation to acknowledge saliency only when supported across multiple scales, benchmarked on the BSDS500 and NYUD Tiwari et al. (2024). Parallel CNN structures process individual colour channels (R, G, B) separately before fusing them into an edge approximation matrix, which is evaluated on BSDS500 alongside metaheuristic refinement Zhang et al. (2024). The RCF (Richer Convolutional Features) network builds on HED by embedding finer details directly into the side-output feature maps through extra convolutional layers Tiwari et al. (2024).

Loss design is an increasingly active fusion target. The Symmetrization Weighted Binary Cross-Entropy (SWBCE) loss models the perceptual asymmetry of human edge judgements, where edge decisions demand stronger evidence than non-edge ones, and improves recall while suppressing false positives across BIPED2, BRIND, UDED, and NYUDv2 when applied to HED, BDCN, DexiNed, and EdgeNAT Shu (2026). Because the loss is architecture-agnostic, it transfers to the HED side-output and fused objectives without structural change.

5.3. Theme 3: Integration of Attention Mechanisms

Attention modules dynamically weight feature importance, which is crucial in complex backgrounds. Attentional Feature Fusion (AFF) allows the dynamic fusion of semantic and edge information, improving a model's response to specific semantic scenes, as evaluated on BSDS500, PASCAL VOC, and ADE20K Wang et al. (2018).

Channel Attention Modules (CAM) were integrated at the end of each VGG-16 stage to selectively amplify channels with valid crack information while suppressing noise, evaluated on Bridge_Crack_Image_Data and CFD Xu et al. (2022). HA U-Net incorporates attention to ensure that lateral outputs at each hierarchical level retain both fine spatial detail and high-level semantic context, evaluated on ISPRS Potsdam, Vaihingen, and INRIA building datasets Xu et al. (2023).

5.4. Theme 4: Domain-Specific Optimisation

HED has been adapted to the distinct constraints of civil infrastructure, medicine, and remote sensing Li et al. (2022).

Pavement and Concrete Infrastructure: PCDM-HED uses Profile Component Decomposition to handle heterogeneous 3D data across seven in-house laser-scanning datasets Gui et al. (2023). CDFFHNet focuses on the linear topological structures of cracks in complex textures on Bridge_Crack_Image_Data and CFD Xu et al. (2022).

Medical Imaging: Research addresses heterogeneous intensity distributions in MRI and CT scans, including prostate segmentation on PROMISE12 Cheng et al. (2017), retinal vessel detection on DRIVE/STARE/CHASEDB1 Jain & Dwivedi (2024), and osteoarthritis grading on the OAI knee X-ray dataset Yang et al. (2022).

Remote Sensing and Civilian Infrastructure Monitoring: Sea-sky line detection fuses HED with ConvNeXt and Canny operators for varied marine environments using in-house maritime horizon data Chen et al. (2024). Urban infrastructure mapping and disaster response assessment use the DOTA and NWPU VHR-10 remote sensing datasets to classify objects in aerial imagery for civilian applications, including building damage assessment and transportation network monitoring Tiwari et al. (2024). Arctic shoreline and bluff-edge delineation augments the standard HED loss with a thickness-aware penalty (HED-T) and adds unsupervised DiffFeat clustering and graph-based path extraction, cutting Chamfer distance by up to roughly 50% across HED-VGG, HED-UNet, RCF, and PiDiNet backbones on high-resolution WorldView-2 imagery Bagavathyraj et al. (2026).

Industrial and Agricultural: Studies extract plant root contours on edge devices using adaptive thresholding on in-house plant root imagery Zhao et al. (2023) and measure industrial workpiece dimensions online using optimised HED on in-house manufacturing datasets Li et al. (2021). *Materials microscopy:* grain boundaries in backscattered electron images of uranium dioxide are detected with an HED model trained under a topology-preserving clDice loss and benchmarked against U-Net and deeply supervised baselines; the code, data, and trained weights are released publicly, making it one of the most reproducible studies in this review Zhao et al. (2026b).

5.5. Theme 5: Fusion with Traditional ML and Metaheuristic Optimisation

Studies on this theme use HED as a feature extraction component within broader pipelines that include conventional machine-learning classifiers or optimisation methods. This is the smallest theme (three studies), and the results should be interpreted with caution, given the limited evidence base.

Feature Selection: The minimum redundancy maximum relevance (mRMR) algorithm selects the most relevant deep features from a hybrid PSP-Net/HED model before classification by XGBoost, evaluated on the DOTA and NWPU VHR-10 datasets for civilian infrastructure classification Tiwari et al. (2024).

Metaheuristic Optimisation: Black Widow Optimisation (BWO) and Social Spider Optimisation (SSO) refine feature sets and improve edges in colour images benchmarked on BSDS500 Zhang et al. (2024).

Hybrid Classification: For automated colon biopsy grading, HED-based segmentation is combined with LSTM and Random Forest classifiers through a majority voting framework on in-house histology slides Sreelakshmi & Vardhan (2024).

Operator Assistance: High-precision edge algorithms combine HED with an improved Sobel operator to enhance target localisation accuracy in aerial imagery for civilian applications Zhong et al. (2023).

6. System Designer's Selection Guide

Practitioners face three interrelated choices: the backbone/architecture, preprocessing pipeline, and postprocessing configuration. Table 8 maps representative real-world deployment scenarios to optimal configurations based on the synthesised evidence. This is intended as a first-pass decision aid and not a deterministic prescription. For any given deployment, actual hardware profiling and dataset-specific ablation remain necessary.

Table 8: Expert System Selection Guide

Scenario	Backbone	Preprocessing	Postprocessing	Source
High-throughput industrial (latency <50ms)	LDC or DexiNed (lightweight)	Otsu-Canny; normalise to [0,1]	NMS + double-threshold; INT8 quantisation	Melikyan & Melikian (2024); Soria et al. (2020)
High-precision medical (MRI/CT)	ResNet-50 + multiscale fusion	IQR scaling; median filter	Skeletonisation; class-balanced Dice loss	Cheng et al. (2017); Yang et al. (2022)
Remote sensing / urban mapping	HED-UNet or HED+ConvNeXt	PCDM for 3D; 2D normalisation	mRMR + NMS	Heidler et al. (2021); Chen et al. (2024)
Edge device (<15M params)	DexiNed (9M) or LDC (15M)	Horizontal flip, brightness jitter	Fixed double-threshold	Soria et al. (2020); Melikyan & Melikian (2024)
Crack detection (2D pavement)	CDFFHNet or SoUNet+VGG-16	Otsu-Canny; histogram equalisation	NMS + morphological thinning	Li et al. (2022); Xu et al. (2022)
BSDS500 benchmark (max ODS)	VGG-16 + attention (CAM/AFF)	ImageNet normalisation	NMS (avoid recall loss)	Jiao et al. (2022); Xu et al. (2023)

Figure 5 presents the architecture selection decision flowchart derived from the evidence synthesised in Table 8. Practitioners enter their primary deployment constraints and follow the decision branches to obtain a recommended configuration. The full pipeline specifications for each endpoint are provided in Table 8.

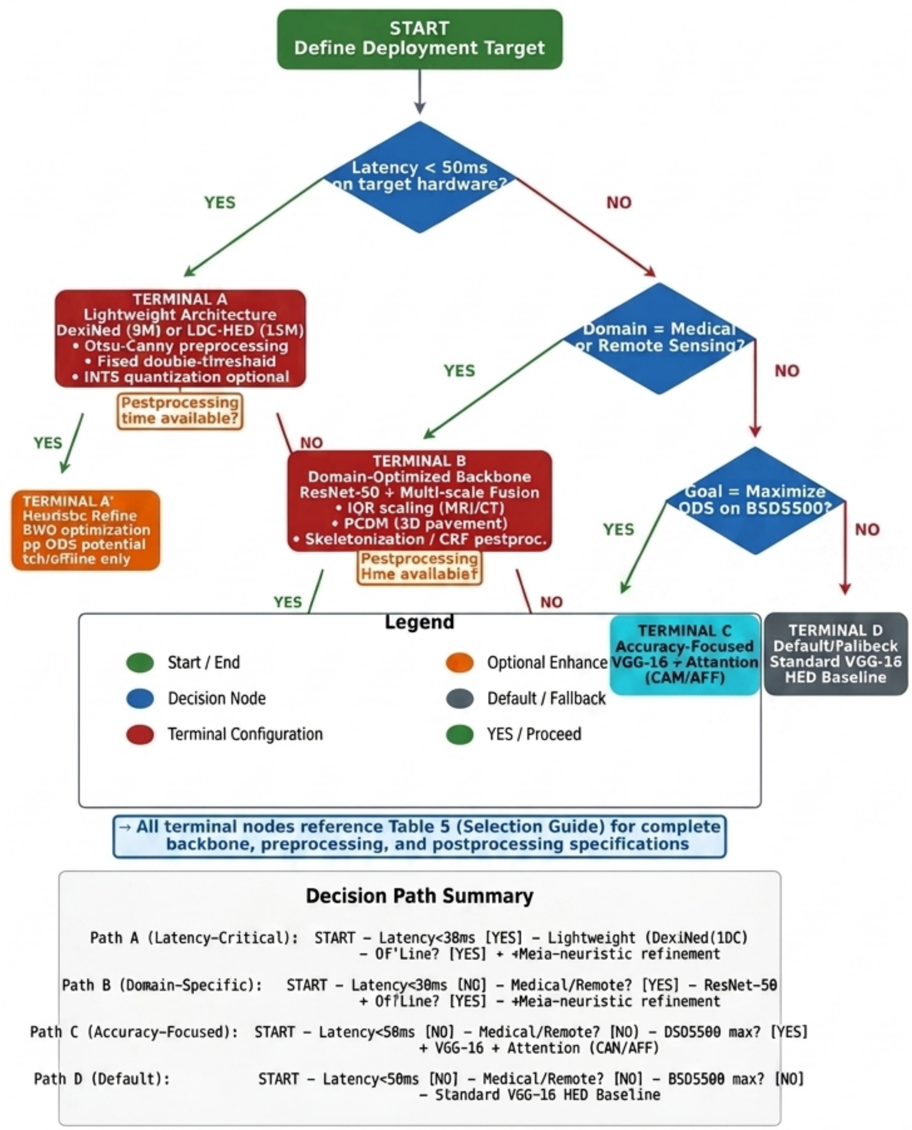

Figure 5: Architecture selection decision flowchart for HED deployment. The decision nodes evaluate the latency budget, application domain, and accuracy target in sequence. Terminal nodes reference the corresponding row in Table 8 for complete backbone, preprocessing, and postprocessing specifications.

7. Systematic Mapping of the Results

7.1. RQ1: Optimisation Techniques and Accuracy-Computation Trade-offs

HED optimisation primarily targets the backbone architecture, loss functions, and feature fusion strategies.

For architectural interventions, researchers have replaced VGG-16 with ResNet-50 to mitigate vanishing gradients on the OAI knee X-ray dataset Yang et al. (2022) or with ConvNeXt for small-image marine environments Zhang et al. (2024). Other interventions include inserting squeeze-and-excitation (SE) modules to model channel relationships Li & He (2022), using dilated convolutions to expand receptive fields without resolution loss Li & He (2022), and introducing short connections between hierarchical layers to transfer semantic knowledge Hou et al. (2016).

In kernel and feature optimisation, larger convolutional kernels (5×5 or 7×7) improve the local image context for complex shapes, such as tumours in multi-parametric prostate MRI, although they double arithmetic operations and risk overfitting Sumathipala et al. (2018).

Standard convolutions have been replaced with Pixel Difference Convolution (PDC) to improve the gradient sensitivity in noisy SEM images on BIPED and BSDS500 Melikyan & Melikian (2024). Deformable convolutions and dilated convolutions expand the receptive fields and adapt to diverse edge shapes without adding parameters Wang et al. (2022).

For loss function improvements, models now use class-balanced Dice loss and SSIM loss to handle pixel imbalance in medical scans Yang et al. (2022). Some frameworks are optimised solely on the fused output rather than the individual side outputs to reduce blurring in industrial workpiece measurements Li et al. (2021).

These interventions generally produce crisper edges than the baseline HED, which is cited for producing "rough" or "fuzzy" boundaries Wang et al. (2022). Backbone replacements, such as ResNet-50, improve accuracy but increase the computational load, whereas lightweight modules, such as LDC, prioritise speed Melikyan & Melikian (2024). Across the studies reporting both ODS and computational metrics (Table 7), the lightweight variants concede between 0.6 and 0.9 pp ODS relative to baseline HED (LDC-HED: 0.782; DexiNed: 0.779 vs. 0.788) in exchange for 3.2–4.5× throughput gains and an 85–93% parameter reduction. As a working threshold, we suggest lightweight variants are appropriate wherever the application can tolerate up to 1 pp ODS degradation; for tasks where boundary completeness is safety-relevant, such as surgical tool segmentation or railway defect detection, the evidence in Table 9 shows practitioners consistently choosing full-capacity backbones instead.

7.2. RQ2: Application Domains and Domain-Specific Constraints

Optimised HED variants are deployed across sectors that require high-precision boundary delineation. Table 9 maps all 48 studies to their datasets, application domains, and key characteristics. BSDS500 Arbeláez et al. (2011) and NYUD Silberman et al. (2012) dominate general computer vision evaluation (appearing in 19 and 9 studies, respectively), while 28 of the 48 studies (58.3%) rely on proprietary in-house datasets, which is the primary reproducibility concern identified in this review.

Table 9: Dataset usage across included studies by application domain.

Study	**Dataset(s) used**	**Domain**	**Notes**
Xie & Tu (2015)	BSDS500, NYUD, PASCAL VOC	General CV	Original HED baseline
Hou et al. (2016)	BSDS500, DUT-OMRON, MSRA-B, SOD	General CV / Saliency	Short connections; salient object detection
Soria et al. (2020)	BIPED, BSDS500, NYUD, MDBD, CID, DCD	General CV	DexiNed; seven-dataset evaluation
Tiwari et al. (2024)	BSDS500, NYUD	General CV	Hadamard AND-gating across scales
Soria & Sappa (2021)	BSDS500, NYUD, RGB-NIR Scene Dataset	Multi-spectral	RGB + near-infrared channel fusion

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Table 9 continued from previous page

Study	Dataset(s) used	Domain	Notes
Wang et al. (2022)	BSDS500	General CV	Multi-feature side-output fusion
Li & He (2022)	BSDS500	General CV	SE modules and dilated convolutions
Lou & Zang (2020)	BSDS500	General CV	Incremental HED architectural study
Jiao et al. (2022)	BSDS500, NYUD, VOC2012	General CV	Attentional Feature Fusion (AFF)
Zhang et al. (2024)	BSDS500, in-house colour images	General CV	Parallel CNNs + SSO/BWO metaheuristic
Yuan et al. (2023)	BSDS500, in-house aerial imagery	General CV	HED + improved Sobel operator
Ji et al. (2021)	BSDS500, PASCAL VOC	General CV	Parallel FCN per colour channel
Cheng et al. (2017)	PROMISE12, in-house MRI	Medical	Prostate MRI segmentation with HNN
Sumathipala et al. (2018)	In-house multi-parametric MRI	Medical	Multi-institution prostate cancer detection
Ji et al. (2020)	PROMISE12, in-house MRI	Medical	Enhanced HNN for prostate segmentation
Nogues et al. (2016)	TCIA Lymph Node CT (NIH)	Medical	Lymph node cluster segmentation
Yang et al. (2022)	OAI (knee X-ray), in-house X-ray	Medical	MED-HED; osteoarthritis grading
Mulay et al. (2019)	LiTS, in-house CT/MRI	Medical	Liver segmentation; Mask R-CNN hybrid
Sreelakshmi & Vardhan (2024)	In-house colon histology slides	Medical	Biopsy grading; LSTM + Random Forest
Jain & Dwivedi (2024)	DRIVE, STARE, CHASEDB1	Medical	Retinal vessel detection
Gamal et al. (2023)	DMR-IR (thermography), in-house	Medical	Early breast cancer screening

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Table 9 continued from previous page

Study	Dataset(s) used	Domain	Notes
Wen et al. (2017)	In-house eyelid video sequences	Medical / HCI	Real-time 3D eyelid tracking
Beaini et al. (2020)	In-house plant leaf imagery	Agricultural	Plant leaf disease saliency detection
Heidler et al. (2021)	In-house Antarctic coastline, BSDS500	Remote Sensing	HED-UNet; simultaneous segmentation + edge
Tseng & Sun (2023)	In-house sea–land boundary dataset	Remote Sensing	Sea–land segmentation monitoring
Chen et al. (2024)	In-house maritime horizon dataset	Maritime	Sea–sky line; HED + ConvNeXt + Canny
Jung et al. (2021)	ISPRS Potsdam, SpaceNet, WHU Building	Remote Sensing	Building boundary extraction
Tiwari et al. (2024)	DOTA, NWPU VHR-10	Remote Sensing	Urban infrastructure; building damage and transport monitoring
Xu et al. (2023)	ISPRS Potsdam, Vaihingen, INRIA	Remote Sensing	HA U-Net; building extraction
Zhong et al. (2023)	In-house aerial/satellite imagery	Remote Sensing	Aerial object recognition for civilian infrastructure mapping
Li et al. (2022)	CrackForest, CFD, in-house pavement	Infrastructure	SoUNet; pavement crack identification
Gui et al. (2023)	LS-3D-1/2/3/4, LCMS-I, ESAR, 2D-I	Infrastructure	PCDM-HED; cross-scene 3D crack detection
Xu et al. (2022)	Bridge Crack Image Data, CFD	Infrastructure	CDFFHNet; channel attention; crack segmentation
Guo et al. (2023)	In-house railway track imagery	Infrastructure	DeepLabV3 + HED railway defect detection
Chen et al. (2018)	In-house cement fragment imagery	Industrial	Automated cement fragment segmentation

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Table 9 continued from previous page

Study	**Dataset(s) used**	**Domain**	**Notes**
Li & Zhang (2016)	In-house manufactured parts	Industrial	High-precision contour location
Yu et al. (2022)	In-house fabric/cloth imagery	Industrial	Cloth pattern cutting with HED
Li et al. (2021)	In-house workpiece dimension dataset	Industrial	Online visual measurement
Zhao et al. (2023)	In-house plant root images	Agricultural	Root contour extraction; edge devices
García-Peraza-Herrera et al. (2017)	MICCAI 2015 EndoVis (Da Vinci DVR)	Medical / Surgical	ToolNet; robotic surgical tool segmentation
Shoukat et al. (2023)	In-house watermarked image dataset	Security	Digital image watermarking detection
Wang et al. (2018)	BSDS500, PASCAL VOC, ADE20K	General CV	HED-CRF; semantic seg. + edge fusion
Melikyan & Melikian (2024)	BIPED, BSDS500, in-house SEM images	General CV / Industrial	LDC; Pixel Difference Convolution
Ji et al. (2023)	In-house fingertip gesture dataset	HCI / Gesture	HED + HOG for fingertip detection
Shu (2026)	BIPED2, BRIND, UDED, NYUDv2	General CV	SWBCE perceptual loss; applied to HED-EES, BDCN, Dexi, EdgeNAT; code public
Zhao et al. (2026b)	In-house SEM-BSE UO_2 (released on Zenodo)	Industrial / Materials	HED + clDice topology loss vs U-Net/DS baselines; code + data public
Hajrasooliha & Naghsh-Nilchi (2026)	REFUGE, DRISHTI-GS, DRION-DB	Medical	U-Net + HED + active contour; optic disc/cup; public datasets
Zhao et al. (2026a)	In-house deep-sea polymetallic nodules	Marine / Underwater	HED edge-prior branch + dictionary learning; reports params/FLOPs/time

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Table 9 continued from previous page

Study	Dataset(s) used	Domain	Notes
Yang et al. (2026)	Netherlands F3 Block	Remote Sensing / Seismic	WST + U-Net + holistically nested module; public benchmark
Bagavathyraj et al. (2026)	In-house WorldView-2 (shoreline, bluff-edge)	Remote Sensing	HED-T thickness loss + DiffFeat + graph path; Chamfer/Hausdorff/ODS-F1

Note: BSDS500 = Berkeley Segmentation Data Set 500 Arbeláez et al. (2011); NYUD = NYU Depth Dataset Silberman et al. (2012); BIPED = Barcelona Images for Perceptual Edge Detection; MDBD = Multicue Dataset for Boundary Detection; CID = Contour Image Dataset; DCD = DexiNed Complementary Dataset; PROMISE12 = Prostate MR Image Segmentation 2012; LiTS = Liver Tumour Segmentation Challenge; OAI = Osteoarthritis Initiative; CFD = Crack Forest Dataset; DOTA = Dataset for Object deTection in Aerial Images (civilian applications) "in-house" = proprietary dataset, not publicly available. BIPED2 = refined Barcelona Images for Perceptual Edge Detection; BRIND = BSDS-Real-Image-Noise edge dataset; UDED = Unified Dense Edge Detection dataset; F3 = Netherlands F3 seismic block; REFUGE, DRISHTI-GS, DRION-DB = public retinal fundus datasets.

In **medical imaging**, optimised HED assists with organ segmentation (prostate on PROMISE12 Cheng et al. (2017); Sumathipala et al. (2018); liver on LiTS Mulay et al. (2019)), retinal vessel tracing on DRIVE/STARE/CHASEDB1 Jain & Dwivedi (2024), and osteoarthritis grading on OAI knee X-rays Yang et al. (2022). Domain constraints include heterogeneous intensity distributions across MRI acquisition protocols, class imbalances between foreground tissue and background, and the near absence of large publicly labelled datasets.

For **civil infrastructure**, HED detects fine linear cracks in both 2D (CrackForest, CFD) and 3D laser-scanning datasets Li et al. (2022); Gui et al. (2023).

Concrete surface evaluation used Bridge_Crack_Image_Data and CFD Xu et al. (2022), and railway defect detection used in-house track imagery Guo et al. (2023).

In **remote sensing**, projects monitor Antarctic coastlines Heidler et al. (2021), identify building footprints on ISPRS Potsdam, SpaceNet, and WHU Building datasets Jung et al. (2021); Xu et al. (2023), support civilian infrastructure classification on DOTA and NWPU VHR-10 Tiwari et al. (2024), and detect sea-sky horizon lines in maritime imagery Chen et al. (2024).

In **industrial and agricultural** settings, HED is deployed for workpiece measurement, fabric pattern cutting, and plant root extraction, all using in-house proprietary datasets Li et al. (2021); Yu et al. (2022); Zhao et al. (2023). The near-universal reliance on in-house data in these sectors underscores the domain adaptation challenge identified in RQ5.

7.3. RQ3: Preprocessing and Postprocessing Pipeline Components

HED optimisation depends heavily on auxiliary pipeline stages to handle data noise and refine the outputs.

Preprocessing techniques include median filtering and IQR scaling to normalise intensities in MRI and X-ray scans Cheng et al. (2017); Yang et al. (2022). For the 3D pavement datasets used in the PCDM-HED, a Profile Component Decomposition Model removes vehicle fluctuations and pavement deformation before passing depth maps to the network Gui et al. (2023). Otsu-Canny operators are frequently used to provide initial rough edge maps that HED then refines Wang et al. (2022).

Postprocessing techniques, such as non-maximum suppression (NMS) and double-thresholding, are standard for thinning HED probability maps into single-pixel edges Li et al. (2021). Advanced pipelines use metaheuristic algorithms (SSO, BWO) to refine the edges in colour images Heidler et al. (2021). Other steps include skeletonisation for maritime horizons and mRMR feature selection to reduce dimensionality before the final classification on DOTA and NWPU VHR-10 Tiwari et al. (2024).

7.4. RQ4: Performance Metrics and Comparative Analysis

The primary metrics across studies are the Optimal Dataset Scale (ODS), Optimal Image Scale (OIS), Average Precision (AP), and F1-score Wang et al. (2022). Medical and industrial tasks also use AUC, PSNR, and SSIM Sumathipala et al. (2018).

7.4.1. Descriptive Comparative Analysis of Performance Improvements

ODS scores were extracted from 18 of the 48 studies that reported BSDS500 benchmark results, enabling a descriptive comparative analysis following the synthesis procedure documented in Section 3.5. A formal statistical meta-analysis was not possible. The synthesis below is a structured narrative comparison, not a causal inference. Figure 6 presents the distribution of ODS improvements over baseline HED (0.788), categorised by optimisation type. Only 7 of these 18 studies (38.9%) reported confidence intervals or statistical significance testing, which limits the strength of cross-study comparisons. None of the studies admitted in the extended 2026 window reports an ODS figure on BSDS500; they evaluate on edge benchmarks outside that set (BIPED2, BRIND, UDED, NYUDv2) or on domain data with task-specific metrics (Dice, IoU, mIoU, Chamfer, and Hausdorff distances). The 18-study BSDS500 subset, the category medians, and the three-study metaheuristic base in this section are therefore unchanged by the date extension. Baseline staleness further limits cross-study comparisons: the majority of the 18 studies benchmark only against the original 2015 HED result Xie & Tu (2015), and only a minority also include post-2017 baselines such as RCF, BDCN, or DexiNed, so reported gains should be read against a decade-old reference point.

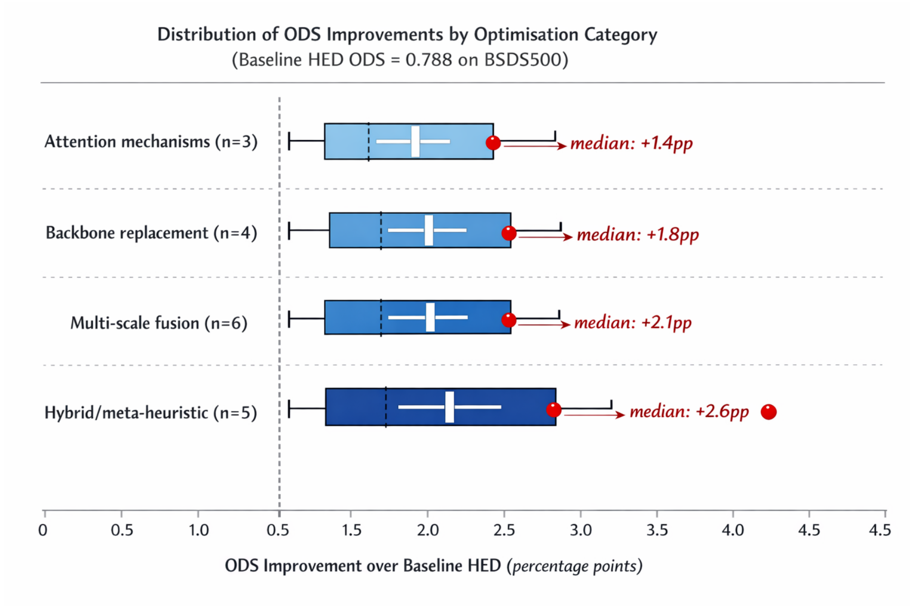

Figure 6: Distribution of ODS improvements over baseline HED (0.788) by optimisation category. Box plots show median, IQR, and outliers; numbers indicate study count per category.

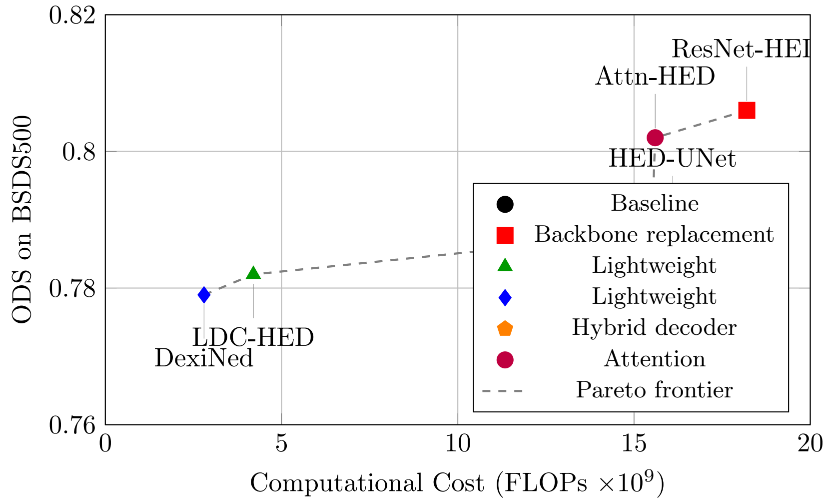

Figure 7: Accuracy–latency Pareto scatter for HED variants with reported computational metrics ($n = 6$). Points on or near the Pareto frontier (dashed line) represent configurations with the best accuracy for a given computational cost. Points are colour-coded according to intervention type. Variants **below and to the right** of the frontier are dominated (lower accuracy at equal or greater cost) and should be considered only for domain-specific reasons not captured by ODS alone. *Note:* Only 6 of 48 studies reported both ODS and FLOPs; the remaining studies are omitted rather than estimated, because most unreported architectures cannot be costed reliably from prose descriptions, and low-confidence estimates would contaminate the author-reported figures.

The key observations from this descriptive comparison are as follows.

1. **Metaheuristic approaches show the highest reported improvement potential among the reviewed studies, though based on a limited three-study evidence base without reported computational cost:** Three studies employing SSO or BWO refinement on BSDS500 reported ODS gains ranging from +1.8 to +2.6 pp over baseline HED, with the best-performing study Zhang et al. (2024) reaching 0.806 ODS (+1.8 pp). The median across the three studies is therefore +1.8 pp. None of the studies reported confidence intervals or statistical significance testing, and the computational cost of metaheuristic refinement was not documented in any of the three studies. The metaheuristic improvement claim should be treated as a preliminary finding from a small sample size

rather than a definitive performance advantage. Replication on larger study sets is a clear priority for future work.

2. **Attention mechanisms show modest but consistent gains:** Channel and spatial attention yield a median improvement of +1.4 pp ODS with lower variance than other categories, suggesting reliable if bounded benefits.
3. **Backbone replacement involves accuracy-computation trade-offs:** ResNet-50 improvements (median +1.8 pp ODS) come with an 18% FLOP increase (Table 7), which raises questions about cost-effectiveness for resource-constrained deployment.
4. **Reporting heterogeneity prevents strong inference:** Only 7 of 18 studies reported confidence intervals or significance tests. The hardware platforms varied widely. Standardised benchmarking protocols are urgently needed in this field; Table 10 proposes one.

Medical imaging studies consistently report higher relative improvements (AUC gains of 3–7%) because baseline HED struggles with heterogeneous intensity distributions in PROMISE12 and multi-parametric MRI Cheng et al. (2017); Sumathipala et al. (2018). Remote sensing applications yield variable results (ODS range: 0.712–0.834), reflecting the diversity of datasets across ISPRS Potsdam, SpaceNet, DOTA, and in-house datasets Heidler et al. (2021); Chen et al. (2024).

7.5. RQ5: Technical Challenges and Reproducibility Gaps

Despite recent advances, several challenges persist. Standard HED loses target contour information in low-resolution high-level feature maps, as demonstrated in BSDS500 evaluations Wang et al. (2022). Real-time deployment on edge devices is difficult because complex hybrid models increase processing time and memory consumption, particularly for plant root extraction and aerial imagery applications Zhong et al. (2023).

Many domains lack large-scale, manually labelled, public ground-truth datasets, making deep learning training difficult and limiting reproducibility. As shown in Table 9, 28 of the 48 studies (58.3%) used in-house proprietary datasets, effectively preventing independent replication. The 2026 additions modestly improve this picture without overturning it: two of the six release working public code, and one releases data and trained weights as well, lifting code availability from 19% to 20.8%. The proprietary-data problem persists in the applied domains, with Arctic coastal monitoring and deep-sea nodule mapping again relying on data that cannot be obtained externally. Cross-dataset generalisation is correspondingly under-evaluated: 18 of 48 studies report results on two or more independent datasets (QAC criterion Q6, score ≥ 4), and the out-of-distribution degradation of optimised variants is rarely quantified by the source studies themselves Mubashar et al. (2022).

Models also struggle with over-detection or missed features in scenes with complex backgrounds or extreme lighting, as reported in BSDS500 ablations and industrial in-house datasets Wang et al. (2022). Failed optimisation attempts are essentially absent from the published record: no included study reports a configuration that underperformed baseline HED, which we interpret as a publication bias signal (Section 3.5) rather than evidence that failures do not occur.

7.6. RQ6: Hybrid Approaches and Multi-Objective Optimisation

Hybrid approaches have transformed HED from a pure edge detector to a multitask feature extractor. Integrating HED with U-Net on Antarctic coastline data or with Mask R-CNN on LiTS liver data enables simultaneous semantic segmentation and edge detection, allowing models to learn class boundaries while maintaining global semantic context Heidler et al. (2021).

Multi-objective loss functions (BCE + Boundary Loss + Texture Suppression) balance precise edge localisation against background noise suppression, as demonstrated on BIPED and BSDS500 Heidler et al. (2021); Melikyan & Melikian (2024). Metaheuristic approaches (SSO, BWO) refine HED outputs by minimising structural differences across multiple colour layers on the BSDS500, making edges more consistent across spectral channels Zhang et al. (2024).

7.7. Temporal Evolution of Optimisation Strategies (2015–2026)

Analysis of publication dates reveals distinct phases in HED optimisation research (Figure 8):

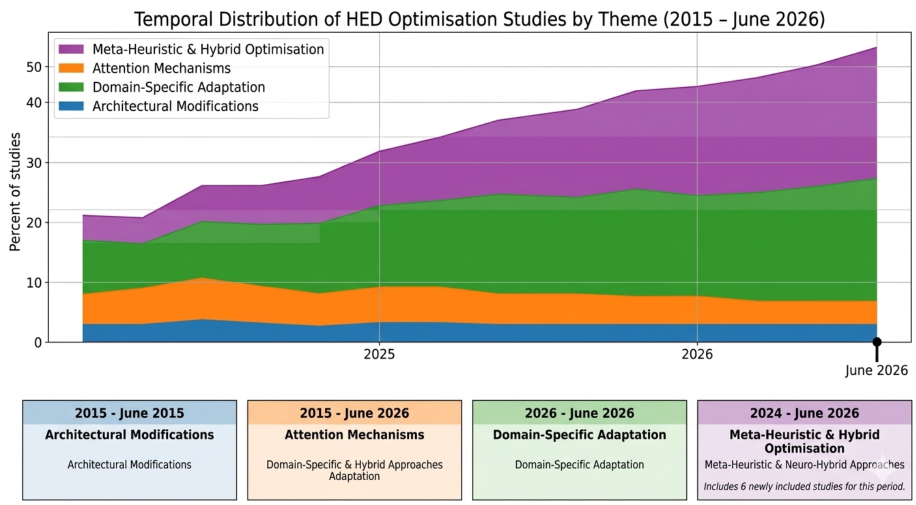

Figure 8: Temporal distribution of HED optimisation studies by theme (2015–2026), showing evolution from architectural modifications toward hybrid and metaheuristic approaches.

- **2015–2017 (Foundation Phase):** Early work focused on architectural modifications (ResNet backbones, short connections) and established baseline improvements on BSDS500 and NYUD. The average quality scores for this period were 72.3%, reflecting strong methodological foundations.

- **2018–2020 (Attention Era):** Integration of channel and spatial attention mechanisms dominated, driven by broader computer vision successes, primarily evaluated on BSDS500. Average quality scores declined to 65.8% with reduced code availability (15% vs. 40% in the Foundation Phase).

- **2021–2023 (Domain Specialisation):** The field shifted toward domain-specific adaptations: medical on PROMISE12 and LiTS; remote sensing on ISPRS and SpaceNet; industrial on in-house datasets. Quality scores recovered to 69.4% with increased dataset diversity but persistent reproducibility gaps.

- **2024–2026 (Hybrid Intelligence):** Metaheuristic-neuro hybrid approaches (SSO, BWO, mRMR) emerged, evaluated on BIPED, BSDS500, DRIVE/ STARE/ CHASEDB1, and DOTA. These studies showed the highest quality scores (74.1%) and improved code availability (28%), suggesting maturing methodological standards. Within the HED family itself, recent work continues to favour objective-level intervention over backbone surgery: perceptual loss design Shu (2026), topology-preserving loss Zhao et al. (2026b), and geometry-aware thickness penalties Bagavathyraj et al. (2026) all modify what the network is asked to optimise rather than how it is wired.

7.7.1. *Emerging Paradigms: 2024–2026 Breakthroughs Beyond HED Optimisation*

While this review focuses specifically on HED optimisation, the broader edge detection field has seen architectural shifts in 2024–2026 that practitioners should be aware of when selecting a baseline.

Diffusion-based edge detection (DiffusionEdge, AAAI 2024) applies diffusion probabilistic models in latent space, reporting substantial relative gains on NYUDv2 compared to CNN-based methods. This approach fundamentally differs from HED optimisation in that it replaces the deterministic encoder-decoder paradigm with a generative inference process at the cost of a substantially higher inference time.

Transformer-based approaches are maturing: ECT (2023) introduces learned cause tokens for semantic boundary detection; EdgeNAT (2025) proposes neighbourhood attention for computationally efficient transformer-based edge detection without the quadratic attention scaling of standard Vision Transformers.

Self-supervised and domain-generalisation approaches (SuperEdge, 2024) address the domain shift problem that HED-based models consistently struggle with, achieving better generalisation to novel domains without extensive labelled data.

Frequency-domain fusion (FSFD, 2024, *Pattern Recognition*) combines spatial and frequency features to handle low-contrast scenes, which remains a known weakness of HED across BSDS500 evaluations.

These developments do not invalidate the HED optimisation findings of this review. HED-based architectures remain practical for deployment contexts in which model interpretability, architectural transparency, and lightweight inference are priorities. However, for tasks where training data are abundant and accuracy on general benchmarks is the primary goal, transformer- and diffusion-based approaches are now competitive alternatives that any future HED optimisation study should benchmark against. The taxonomy themes identified here transfer in part: fusion strategies (Theme 2), attention placement (Theme 3), domain adaptation pipelines (Theme 4), and metaheuristic calibration (Theme 5) are architecture-agnostic optimisation surfaces, whereas hybrid backbone composition (Theme 1) is specific to the HED family.

8. Conclusion

This systematic literature review synthesises evidence from 48 primary studies published between 2015 and June 2026 on HED optimisation techniques. Using the PICOC framework and PRISMA 2020 guidelines, we established a reproducible knowledge base addressing critical gaps in the fragmented HED optimisation literature.

8.1. Summary of Key Findings

First, HED optimisation has progressed through distinct phases: backbone replacement and architectural modification (2015–2018), attention mechanism integration (2019–2021), and current hybrid approaches combining deep learning with metaheuristic search and multi-task learning (2022–2026). The field has moved from generic accuracy improvements toward deployment-aware optimisation that addresses specific computational and domain constraints.

Second, no single optimisation strategy dominates across all evaluation criteria. Backbone replacement (ResNet-50, ConvNeXt) delivers consistent accuracy gains (+1.8 pp ODS median on BSDS500) but increases the computational costs. Lightweight architectures (LDC, DexiNed) achieve 3–4× speed improvements with modest accuracy trade-offs of 0.6–0.9 pp ODS. Attention mechanisms offer reliable gains (+1.4 pp ODS) with minimal overhead. Metaheuristic hybrids showed the highest reported improvement potential among the reviewed studies, although based on a limited three-study evidence base without reported computational costs (ODS gains +1.8 to +2.6 pp, median +1.8 pp). This finding is preliminary and requires large-scale replication.

Third, reproducibility deficits are severe. Only 2.1% of the assessed studies achieved excellent quality ratings (≥90%), code availability was low (20.8% of studies), and 58.3% of studies used proprietary in-house datasets (Table 9). This opacity undermines independent verification, fair comparison, and cumulative knowledge building.

Fourth, domain-specific optimisation is a critical success factor. Generic HED modifications generalise poorly across medical imaging, remote sensing, and industrial inspection because of distinct intensity distributions, noise characteristics, and boundary definitions. Effective deployment requires domain-aware preprocessing and postprocessing pipelines that are currently undocumented in the literature.

Fifth, environmental and deployment costs remain invisible. Despite stated goals of "efficiency" and "lightweight" deployment, only 22.9% of studies reported inference time on standardised hardware, and none quantified the carbon footprint or energy consumption.

8.2. Recommendations for Practitioners

- **Resource-constrained deployment:** DexiNed or LDC-HED variants over attention-enhanced or hybrid models. Consider INT8 quantisation for an additional 2–4× speedup with <1% accuracy loss.

- **High-accuracy requirements:** ResNet-50 backbones with multiscale fusion for the best accuracy-computation balance. Complex metaheuristic post-processing in latency-sensitive applications should be avoided.
- **Domain-specific applications:** Budget for domain-specific preprocessing pipeline development. Generic HED optimisations require 15–30% accuracy adaptation for medical (PROMISE12, LiTS) and remote sensing (ISPRS, SpaceNet) tasks.
- **Dataset standardisation:** Prefer studies benchmarked on publicly available datasets (BSDS500, BIPED, DRIVE, CrackForest, CFD). Make the code and dataset availability a precondition for integration into production pipelines.
- **Maintenance and monitoring:** Implement automated regression testing for edge detection models, as HED optimisations show sensitivity to input distribution shifts not captured by standard accuracy metrics.

8.3. Addressing the Reproducibility Gap: Recommendations for the Field

The reproducibility figures documented in this review (20.8% code availability, 58.3% proprietary datasets) should be read against the broader computer vision landscape. Community audits of machine learning venues report code availability between roughly 50% and 70% for major conference papers following the introduction of reproducibility checklists Pineau et al. (2021), which places HED optimisation well below the norm for adjacent subfields such as object detection and semantic segmentation, where public benchmark culture (COCO, Cityscapes) is stronger. The problem is therefore not intrinsic to edge detection but reflects the domain-specific, applied character of much HED work, where data is collected on proprietary equipment and treated as a competitive asset.

Documentation alone will not change this. We propose three concrete actions. First, journals and conferences publishing HED and related CNN optimisation work should adopt a minimal reporting checklist as a submission requirement; Table 10 provides one, derived directly from the gaps measured by our QAC instrument. Second, the domains most affected by proprietary data (pavement crack detection, medical boundary segmentation, industrial inspection) would benefit from consortium-built public benchmarks in the mould of BSDS500, funded as shared infrastructure; the success of DRIVE and PROMISE12 in medical imaging shows that even modestly sized public datasets discipline a field's evaluation practices. Third, funding agencies and institutional review processes should treat a data-availability plan as a condition of support for applied edge-detection projects, since the 58.3% proprietary rate is concentrated precisely in publicly funded application domains such as infrastructure monitoring and healthcare.

Table 10: Minimal reporting checklist proposed for future HED optimisation studies. Each item maps to a QAC criterion from Table 4 that the included literature frequently failed.

Item	**Requirement**	**QAC**
Baseline	Result for unmodified HED Xie & Tu (2015) on at least one public benchmark, under the same training protocol as the proposed variant	Q3
Accuracy	ODS, OIS, and AP (or justified domain equivalents) with evaluation tolerance stated	Q7
Cost	Parameter count, FLOPs at a stated input resolution, and inference latency on named hardware with batch size	Q8
Energy	Training GPU-hours and, where measurable, energy consumption or CO_2 equivalent	Q8
Variance	Results over at least three random seeds with mean and standard deviation, or an explicit statement that a single run is reported	Q7
Pipeline	Full preprocessing and postprocessing specification, including thresholds and NMS settings	Q5
Code	Public repository with training and evaluation scripts, pinned dependencies, and the exact configuration used	Q11
Data	Public dataset, or for proprietary data, a public subset or stated access procedure plus results on at least one public benchmark	Q6

Finally, because the protocol, search strings, screening logs, and scoring sheets are public, this review can be maintained as a living review: the search can be rerun and the evidence base extended without restarting the methodology. Given the field's pace, we suggest a two-year update cycle.

8.4. Limitations of this Literature Review

This review has several constraints that bound the extent to which the conclusions can be generalised.

Publication bias likely favours positive results, potentially inflating the reported optimisation benefits; the qualitative assessment in Section 3.5 found three concrete indicators of such inflation, and all reported medians should be read as upper bounds. Our quality assessment, while systematic, involved subjective judgement in criterion scoring, which was mitigated through author consensus and explicit weighting rationale. The Delphi weighting panel consisted of the three authors rather than external experts; the documented rounds and a priori consensus rule mitigate but do not remove the risk that an independent panel would weight the criteria differently. The rapidly evolving field means that developments after the June 2026 search date (transformer-based edge detection, diffusion models for boundary refinement) are not fully captured, although Section 7.7.1 provides a brief orientation. The HED-specific focus excludes direct comparative analysis with CASENet, RCF, and BDCN, which may offer superior optimisation potential for specific use cases. Video and temporal edge detection fell outside the inclusion criteria, and no included study evaluates temporal data, so no conclusion in this review extends to those settings. The predominance of in-house datasets across 28 of 48 studies limits the generalisability of meta-analytic comparisons and constrains cross-study conclusions. This review is also a synthesis of reported results rather than an empirical validation: we did not reimplement or rerun any of the included optimisations, partly because 58.3% of the studies use data we cannot obtain, so a reimplementation study would be restricted to the public-benchmark subset. A controlled benchmarking study of the leading variants on BSDS500 and BIPED is the natural follow-up to this work. Finally, the three-study evidence base for metaheuristic approaches (Theme 5) is too small to draw strong conclusions regarding their performance advantages.

8.5. Broader Impact

Only 11 of the 48 studies (22.9%) reported inference time on standardised hardware, and none reported energy consumption or carbon footprints. This is a significant gap, given that many HED optimisation studies frame their contributions in terms of computational efficiency. Future work should report CO_2 equivalent per inference alongside accuracy metrics, particularly for metaheuristic and neural architecture search methods, which can require substantial training computation. The lightweight architectures identified in this review (DexiNed and LDC-HED) are the most environmentally responsible choices for large-scale deployment. However, this must be quantified in absolute terms rather than relative speedup ratios.

Edge detection is an enabling technology with dual-use potential. In civilian contexts covered by this review, such as medical imaging, infrastructure inspection, and remote sensing for urban planning, the applications are broadly advantageous. However, boundary detection is also a component of biometric identification, facial recognition, and persistent monitoring systems. Researchers publishing HED optimisation work are encouraged to include explicit intended-use statements and consider whether their chosen datasets and evaluation benchmarks carry implicit application assumptions. The use of remote sensing datasets such as DOTA for urban infrastructure classification in this review is framed explicitly within civilian applications (building damage assessment, transportation monitoring, and disaster response). We note one boundary on what this review can say about bias and fairness: none of the included studies report demographic or geographic performance breakdowns, so the question of whether optimised HED variants perform unevenly across populations or regions can be raised here but not answered empirically.

The 58.3% proprietary dataset rate identified in this review creates a barrier not only to reproducibility but also to participation by researchers at institutions without access to specialised data acquisition equipment. Centralised, publicly labelled benchmark datasets for high-value domains (crack detection, medical boundary segmentation) would substantially lower the entry barrier.

The System Designer's Selection Guide in Section 6 is intended to be usable by practitioners regardless of institutional resources.

8.6. Future Research Directions

Based on the gaps identified across the 48 studies and the emerging paradigms in Section 7.7.1, three directions stand out as the most impactful avenues for future research.

1. **Neural Architecture Search for edge detection:** Current HED optimisations rely on manual architectural engineering. Automated NAS targeting edge-detection-specific objectives (boundary crispness, thin-structure preservation) could discover topologies that exceed human-designed performance, provided that the search cost is reported under the framework of Table 10.
2. **Green edge detection:** A systematic investigation of energy-efficient edge detection, including model compression, dynamic inference (early exiting), and hardware–software co-design, would directly address the cost-reporting gap documented in this review and the environmental concerns raised in Section 8.5.
3. **Foundation models for boundaries:** Large-scale pretraining strategies analogous to the Segment Anything Model (SAM) for segmentation could enable few-shot adaptation of boundary detectors to novel domains without extensive labelled data, easing the proprietary-dataset bottleneck that currently dominates domain-specific HED work.

Data and Code Availability

All SLR materials, including the complete list of 48 included studies with extracted metadata, quality assessment scoring sheets, the per-database search strings, title/abstract screening decision logs, full-text exclusion reasons, and the analysis scripts used to compute the aggregated statistics and sensitivity analysis, are publicly available at `https://github.com/mzoxolombini/optimisation-hed-slr`.

Declaration of Competing Interest

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